

Initial question:

How can artificial intelligence be used to improve the online learning experience?

Summary:

This paper aims to dissect the significance and purpose of education, evaluate how design thinking can be used to build improved learning experiences, and explore ways that machine learning and artificial intelligence can optimize the learning experience.

Table of contents:

Segment 0.1 – INTRODUCTION	(page 2)
SECTION 1 – Purpose, value, and the history of education	
CHAPTER 1 – VALUE	(page 5)
Segment 1.1 – Definitions and Goals	(page 5)
Segment 1.2 – Theoretical Power and Purpose	(page 9)
Segment 1.3 – Environmental Changes and Challenges	(page 13)
CHAPTER 2 – HISTORY	(page 23)
Segment 2.1 – History of education and technology: dawn of man to 1900s	(page 23)
Segment 2.2 – Modern epistemology: 1900s	(page 28)
Segment 2.3 – Digital tools: 2000s	(page 35)
SECTION 2 – Modern design methods and emerging technology	
CHAPTER 3 - DESIGN THINKING	(page 39)
Segment 3.1 – Design Methodologies	(page 40)
Segment 3.2 – Case Study – The Montessori School	(page 44)
CHAPTER 4 - EMERGING TOOLS	(page 47)
Segment 4.1 – Hardware, Software, and Big Data	(page 48)
Segment 4.2 – Interaction Design, Communication, Command	(page 60)
CONCLUSION	
Segment 5.1 – Conclusion	(page 67)
Segment 5.2 – Three Essential Questions	(page 69)
Bibliography	(page 70)

INTRODUCTION

The first half of this research paper begins by looking at how education is essential to humanity and has always been key to the survival of any species. Section one will outline definitions in learning, describe the purpose of education, and observe the greater theoretical powers that education has in society. Section one also explores how human learning has evolved over time, entirely dependent on the evolution of our communication tools. The second half of this paper will review modern design thinking. We examine design methodology and reference a profound case study demonstrating how to apply best practices in successful education design. Finally we will look at emerging digital technologies that are driving today's design innovation and modern communication between computers and humans.

Education matters; everyone seems to agree, but why is learning so important? Is it becoming more or less important with time? There have always been interesting philosophies about education, but increasingly common are the theories that education will save the human species, potentially even from itself!¹ Society's skills and knowledge requirements are continually evolving, and in turn, so too should the goals of our education systems. Given this, what subjects do people need to learn about for the future, and what are the best ways to learn them? In the first chapter of this work we explore the skills and types of knowledge that may best serve (or save) our species based on the current trajectory of our environment. If education is only a tool, will it also be used for harmful purposes too? What are the future threats to humanity and what impact can education have in the best-case scenario, or in the worst? Schools are only a human creation - this means that they are imperfect by design and they will evolve over time with our evolving needs and capabilities. Education is a form of cultural communication, and our educational methods have always evolved within the boundaries of our communication tools. Looking at the history of education in chapter two, the communication-based boundaries of education are made very clear. Since entering the information age, we have continued to see how digital tools can push education into its next generation. Are there still digital functions and capabilities that we haven't yet applied to education design? Absolutely! Digital technology is still relatively young, and advanced digital tools are quite new to education. We are in a time when digital tools, namely smart phones and powerful cloud-based software are providing new

¹ "Most threats to humans come from science and technology, warns Hawking". Guardian, 2016

potential at rapid speeds. In the age of big data, data is the new currency ². Simply put, machine learning is like a new lens we can use to cut through the noise of our current information overload and extract value. It makes more sense with so much big data newly available. Given how machine learning is already widely used to improve other communication services like advertising, social media, media search, and gaming, how might it be used to improve education too?

To have a meaningful discussion about education we review and reference a wide selection of prominent thinkers and consider many of their most significant theories. This education research relies heavily on the ancient philosophies of Plato, scientific hypotheses of Charles Darwin, and concepts developed by renowned psychologists and epistemologists such as Maria Montessori, Jean Piaget, John Dewey, Seymour Papert, and Salman Khan (of Khan Academy). To develop our set of humanitarian priorities we look to the United Nations, The World Economic Forum and other organizations focused on global peace and sustained human development. This research also consults the published opinions of modern science, technology, policy, and innovation leaders like Stephen Hawking, Barack Obama, and Bill Gates. These individuals help to further identify and illuminate the emerging challenges, opportunities and trends that are most likely to impact the human environment.

With the primary goal of adding value in education, we begin our journey with a careful analysis of what defines educational value. What are the educational ideals to work towards? Once we should find out that education is a means of making our species more adaptable, we then explore what many scientists agree to be the greatest obstacles of our future. We will observe the human history of learning and arrive at what recent innovations modern learning tools currently offer. Only after we develop a deep understanding of education and review modern design thinking principles do we finally consider the potential for machine learning. We consider machine learning's dependency on data and try to focus on where opportunities will be for creating value in the education systems of today and tomorrow.

² "Data As The New Currency". Deloitte University Press, 2013

SECTION 1:

Purpose, value, and the history of education

CHAPTER 1 – VALUE

Segment - 1.1 – Definitions & Goals

“Establishing a lasting peace is the work of education;
all politics can do is to keep us out of war.”

- Maria Montessori³ (1949)

To establish a meaningful discussion about education and learning, we begin by laying the foundation with relevant definitions and goals. What is learning? What is knowledge? This section will outline why we must consider the audience, functions, and forms of education tools in order to set goals. What defines an effective teaching tool? This section is intentionally literal, as we will explore philosophies and educational theory deeper in the next section. Please note some basic definitions gathered from the Dictionary.com, Merriam-Webster and Oxford online dictionaries:

- **Education** is “the process of receiving or giving systematic instruction.”⁴
- **Epistemology** is “The theory of knowledge, especially with regard to its methods, validity, and scope, and the distinction between justified belief and opinion.”⁵
- **Information** is “facts or details about a subject”⁶
- **Instruction** is “a statement that describes how to do something”⁷
- **Knowledge** is “information, understanding, or skill gained from experience or education”⁸
- **Learning** is “the activity or process of gaining knowledge or skill by studying, practicing, being taught, or experiencing something.”⁹
- **Process** is “a systematic series of [mechanized or chemical] operations that are performed in

³ Montessori, Education and Peace (1943), Chapter 1, page 21. <PDF - https://archive.org/details/Peace_And_Education_>.

⁴ The Definition Of Education, OxfordDictionaries.com

⁵ The Definition Of Epistemology, OxfordDictionaries.com

⁶ The Definition Of Information, Miriam Webster’s Dictionary

⁷ The Definition Of Instruction, Miriam Webster’s Dictionary

⁸ The Definition Of Knowledge, Miriam Webster’s Dictionary

⁹ The Definition Of Learning, Miriam Webster’s Dictionary - A Britannica company

order to produce something.”¹⁰

- **Skill** is “the ability, coming from one’s knowledge, practice, aptitude, etc., to do something well”¹¹
- **Understanding** is “the knowledge and ability to judge a particular situation or subject”¹²
- **Wisdom** is “knowledge of what is true or right coupled with just judgment as to action; sagacity, discernment, or insight.”¹³

While the terms *education* and *learning* have particularly similar definitions, please note that the ambition of education is to facilitate or even administer learning. Based on the definitions of this section (and for the purpose of this text), education may be defined as a systematic distribution of learning skills, knowledge, and information. Looking at education as a human tool for distributing skills, the success of any education system can be defined by the quality and value of those skills that it transfers to its user-base.

A best selling author and professor, the late Doctor Russell Ackoff (1919-2009) was an American theorist, philosopher and professor described as a “trailblazer in the field of organizational operations, [and] the pre-eminent authority on organizational systems theory.”¹⁴ He has been widely quoted on subject of systems thinking and credited with making the field more understandable. One particularly insightful message of Ackoff states that, “Information, knowledge, and understanding enable us to increase efficiency, not effectiveness... Intelligence is the ability to increase efficiency; wisdom is the ability to increase effectiveness.”¹⁵ Although Ackoff himself did not produce the below graphic, his theories and writings often described the relationships between data, information, knowledge and wisdom conveyed by figure 1-1.

Russell Ackoff’s following words describe well figure 1-1 below;

“An ounce of information is worth a pound of data.

¹⁰ The Definition Of Process, OxfordDictionaries.com

¹¹ The Definition Of Skill, Dictionary.com

¹² The Definition Of Understanding, Miriam Webster’s Dictionary

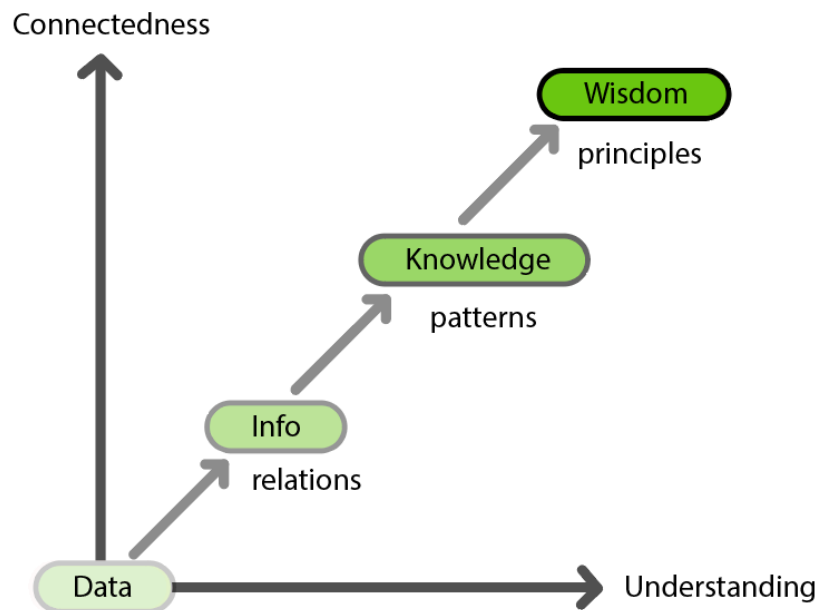
¹³ The Definition Of Wisdom, Dictionary.com

¹⁴ “Ackoff Center Weblog: Russell L. Ackoff, Management Consultant & Systems Thinker, 1919 -2009”

¹⁵ Ackoff, R. L. (1999) *Ackoff’s Best*. New York: John Wiley & Sons, pp 170 – 172. - <http://faculty.ung.edu/kmelton/Documents/DataWisdom.pdf>.

An ounce of knowledge is worth a pound of information.

An ounce of understanding is worth a pound of knowledge.”¹⁶



(Figure 1-1 relationship map of data, information, knowledge, and wisdom¹⁷)

What should be noted from Ackoff's ideas is that attaining wisdom and understanding (the ability to think) are much more valuable than simply gathering knowledge (data and information). This is especially true with the adoption of the Internet, which now allows instant access to virtually limitless information without the need for even basic education.

Thinking further about the literal definition of education tools ("systems for distributing skills"), several criteria should be weighed in determining what makes an education tool more or less successful (as no single element completely defines the success of such a tool). To provide an example, a tool's reach is futile if the content being transmitted is not worthwhile. Imagine if every person on earth should know the exact surface temperature of the sun - it is not clear that this bit of information would serve humanity in any way. On the other hand, the value of educational content is diminished if the distribution method fails to reach its audience properly.

¹⁶ Ackoff, R. L. (1999) *Ackoff's Best*. New York: John Wiley & Sons, pp 170 – 172. op. cit

¹⁷ Original image base on - "Data, Information, Knowledge, & Wisdom". *Systems-thinking.org*. Web. 17 Dec. 2016 . <<http://www.systems-thinking.org/dikw/dikw.htm>>.

In other words, perfect knowledge of relevant information (say a cure for cancer) is still useless if nobody can access it.

What are the goals of any education system? In design thinking, we first take a step back and define the target audience of an education system. We evaluate the audience first to be consistent with the user-centered design approach.¹⁸ The goal then for a general human education is to serve all of humanity. Next, to be deliberate in design, qualitative assessments of educational content must be made. Defining the *best* type of content is challenging (if not impossible) so quality should be evaluated on a few different bases. Assuming first that content will be accurate, it is a good idea to explore what educational functions may be most appropriate. Broadly speaking, the function of any learning is to help the audience adapt to and survive in their changing environment. After finding functional purpose, we can identify which forms and formats should be most appropriate. Forms of education will be bound to communications resources but the ideal medium will deliver the function (adaptability) most effectively. Lastly, we consider how factors for accessibility can be graded, and why they are so important to the success of education tools. Advanced education systems rely on content which is often modular and also consumable at a personalized pace. To summarize, an ideal education system should meet the following criteria: strive to work for the largest possible user-audience, focus on teaching skills that will serve humanity's future interests, and be as freely accessible as possible. Having outlined these literal definitions and the goals of education, we can now consider some prominent theories about the potential powers and purposes which education can serve.

¹⁸ "User-centered Design Basics" per Usability.gov (UK government design guides)

Segment - 1.2 – Theoretical Power and Purpose

“Education is the most powerful
weapon we can use to change the world.”

- (2003) Nelson Mandela¹⁹

Education is an age-old topic with an ocean of opinions. One reason for so much commentary on the subject of learning is that everyone seems to agree it is important. Ancient philosophers and more modern theorists have defended and documented various abilities, challenges, and significances for education. By reviewing some of the most prominent perspectives of all time, we can look beyond the literal definitions and goals (to transfer knowledge and increase human adaptability) of education. Philosophy encourages us to look deeper into *big ideas* about the great purposes that education can serve. Below, we first note how according to Charles Darwin, learning is essential to biological survival. Second we consider how Plato proposed that learning is innate in life and has mind-altering powers.

Looking at education as a systematic process involved in learning we can spot similarities to the concept of natural selection as presented by Charles Darwin in his book, *On The Origin of Species*. In developing his book, Darwin famously spent more than fifteen years documenting many carefully executed scientific tests that ultimately describe a systematic process for sustaining life²⁰ (Page 95). Darwin’s findings eventually focus largely on one truly essential characteristic of sustained life - adaptability. His experiments testify harshly to a theory that if any life form cannot adapt to its environment, it will quickly face extinction. Darwin was not the first to bring up the ominous importance of adaptability but his scientific approach gave more meaning to the philosophy. A sub-theme in Darwin’s study also holds that increased diversity of cells will improve a greater organism’s adaptability and therefore improve its odds of survival. This logic holds that with more variations, a species has more shots at adaptation and ultimately an improved opportunity for survival. This has been echoed since by many scholars and applies to all living organisms, as small as individual germ cells and as large as the human species. To

¹⁹ Nelson Mandela during an address at the Planetarium, Johannesburg, South Africa, 16 July 2003

²⁰ Darwin, Charles. *On The Origin Of Species By Means Of Natural Selection*, 1 October 1859. Page 95.

summarize Darwin's theories, the survival of our species will absolutely depend on our ability to be adaptable, which may also be linked to the diversity of our species²¹.

Roughly 2,500 years ago, in his written work titled *Republica* (514a–520a), Plato is known for some of the earliest philosophies available on learning, education, and human knowledge.²² Plato's famous play, *The Allegory of the Cave*, is a story told by Socrates, another ancient philosopher known as being Plato's teacher. To summarize this play, Socrates narrates the story of three men locked facing a wall in a dark cave where their entire existence is based on simple shadows cast into the cave from unfamiliar objects passing by the obscured entrance to the cave. One of the three men is unchained and dragged from the cave to where he is anxiously blinded by the sun, and slowly adjusts to his first exposure to daylight. When he returns to the cave his perception of the shadows cast onto the wall is forever changed. He now understands that those shadows in the cave are merely cast by the real objects, which truly exist out in the sunlight. He tries explaining his new understanding in a stubborn debate, but the other men refuse to agree even denying his offer to show them.

In the opening lines of Plato's story Socrates asks the reader to;

“compare our nature in respect of education
and its lack to such an experience as this.”

- Socrates²³

Socrates is proposing that we think of education in terms of what our life (in nature) would absolutely be without it. This story is an example of abstraction, leaving the play's message open to various interpretations. While the full story offers commentary on few aspects of human nature (1 - What we know shapes our perspective of life. 2 - What we do not know is often difficult, and even painful to understand. 3 - Once we know something, we do not give up the new perspective easily.) The most significant take-away and general theme throughout is that exploring the unknown, and in turn educating ourselves, can give our lives new meaning. The

²¹ Darwin, Charles. *On The Origin Of Species By Means Of Natural Selection* op. cit., Paraphrasing

²² “The Allegory Of The Cave By Plato: Summary, Analysis & Explanation - Video & Lesson Transcript

²³ Plato, *The Allegory Of The Cave*: Translated by Thomas Sheehan, op. cit., page 1

type of life meaning people may gain from education isn't precise but this story is an observation of the raw power education can have on shaping a person, and their values. In terms of Plato's story, it is through exploration that we access knowledge. In this regard exploration acts synonymously with education. The first cave dweller who is exposed to sunlight goes through a mental transformation after his exploration beyond the cave. Exploration and his discovery of the sun give his life new meaning. This ability to give people's lives new and invigorated meaning is fundamental to the power of education. Thinking also in Darwinian terms, through exploration one cave dweller has adapted mentally while also diversifying the perspectives of his group. His world is altered by his exposure to direct sunlight and he becomes a different type of person, a person likely better suited for the realities of his environment.

Probably the most popularly cited purpose for education today, is the role of education as a means of peace. Per the United Nation's evolving charter, the current mission of the organization is above all else "To maintain international peace and security".²⁴ From the perspective of those committed to peacekeeping and anti-violence, education is a central theme. Given the power of education to transform perspectives and shape human values, it is no wonder then, that the United Nations and most other organizations committed to peace are using education to combat violence and injustice. Peacekeeping and violence prevention seem to be the most commonly cited goals of education today. From The Mahatma Gandhi Institute to the Nobel Peace Prize, and The Mother Teresa Foundation, there is an agreement globally that education gives people the strength to see past violence, overcome poverty, and help the productive development of civilization. The Mother Teresa foundation states that;

"It is because the most terrible poverty in the world is loneliness and the feeling of being unloved. Since its inception, following the footprints of its patron Mother Teresa, The Foundation has been concentrating in the field of education and health of the poor and downtrodden in various ways. The Foundation has been collaborating with other agencies, both governmental and non-governmental towards bringing up the poor children, women and farmers in the mainstream of life."²⁵

How does education create peace? Through out his career, John Dewey (1859-1952), a developmental psychologist completed very in-depth work on this subject. In his 1916 book,

²⁴ United Nations, official website homepage (2016)

²⁵ The Mother Teresa Foundation, official website Welcome page (since 2010)

Democracy and Education: An Introduction to the Philosophy of Education (wherein chapter one is appropriately titled “Education as a Necessity of Life”), Dewey distinguishes that;

“What they [humans] must have in common in order to form a community or society are aims, beliefs, aspirations, knowledge -- a common understanding -- like-mindedness as the sociologists say. Such things cannot be passed physically from one to another, like bricks...”²⁶

Further in his research, Dewey points to a strong link between the three words: common, community, and communication. With special attention to the requirements of forming community, Dewey’s highlights how societal human factors such as proximity and even common goals are not enough to form strong community. His argument is that education itself is what holds people together. It takes shared learning experiences and appropriate levels of communication to understand each other accurately. Dewey’s theories highlight education as a form of universal communication (passed between generations and cultures) that will develop a shared sense of purpose in people. Beyond this, Dewey defines education as the essence of how people can coexist peacefully. His career took a critical look at the power of education and how it must serve mankind in the development of a growing and stable society.

This section has summarized how learning is critical to survival. As the old-age lesson goes, if you do not learn that fire is hot it will burn you. It’s no wonder that education has a tendency to increase a human’s life expectancy²⁷. As Darwin highlighted, a cell must learn to adapt to it’s environment, and as Plato explored thousands of years before Darwin, the power of learning is a most mind-altering and life-changing experience. In the next section we explore what topics it would serve us well to study.

²⁶ Dewey, John. *Democracy and Education*, 1916 (page 5)

²⁷ Harvard Medical School. *Life Expectancy Rises For The Educated; The Less-educated Reap No Benefit*. *Science Daily*, 12 March 2008. <www.sciencedaily.com/releases/2008/03/080311081149.htm>

Segment - 1.3 – Changes & Challenges

"Adapt Or Perish, Now As Ever,
Is Nature's Inexorable Imperative."

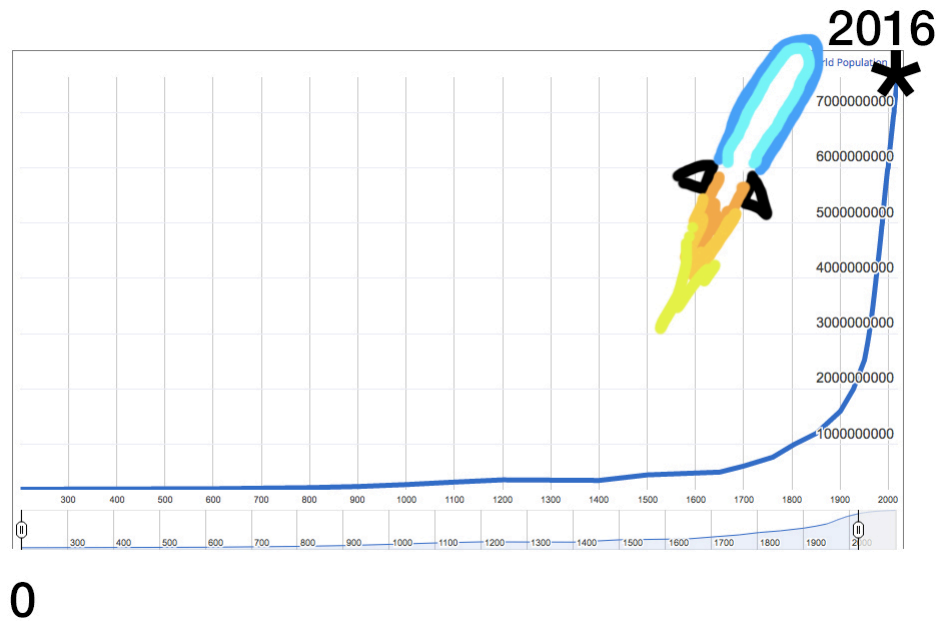
– H.G. Wells (1866-1946)²⁸

Based on Darwin's theories, to what might the human species need to adapt in the future? What changes are we facing in our environment? In fact, shifts caused by our own species represent the greatest changes on the planet. This should come as no surprise given two recent phenomena: the explosion of our population and the exponential growth human technology. A Stanford University organization called Millennium Alliance for Humanity and Biosphere brings together natural scientists, social scientists, and humanists to research what they call "foresight intelligence: the ability to implement behavioral, institutional and cultural changes necessary for humans to ensure a sustainable and equitable future for all." On their website home page they describe the human twenty-first century challenge as "adapting and sizing ourselves to fit within the capacity of one planet."²⁹ The unique characteristic of an exponential growth rate is how it begins with a nearly flat trajectory, making it difficult to detect the curve early on. In more developed stages growth quickly shift into rapid growth with a slope that approaches a vertical line. The population has alarming growth in recent time. In an ongoing global report, The World Economic Forum points out that it took from the dawn of man until 1804 for the global population to reach one billion people. This is in sharp contrast with the most recent billion people to join our population in just the last 12 years (between 2003-2015).³⁰

²⁸ "Adapt Or Perish - Quote - The Free Dictionary. *Forum.thefreedictionary.com*. Web. Nov. 2016 . <http://forum.thefreedictionary.com/postst56991_Adapt-or-perish--now-as-ever--is-nature-s-inexorable-imperative.aspx>.

²⁹ Millennium Alliance for Humanity and Biosphere . N.p., 2016. Web. 11 Dec. 2016 . <[http:// mahb.stanford.edu/blog/the-challenge-for-humanity/](http://mahb.stanford.edu/blog/the-challenge-for-humanity/)>.

³⁰ Worldometers.info/world-population, Population past and present (2016)



(Figure 1-3, population growth from year one A.D. to 2016)³¹

Due to constant technological advancements in health and medicine, births are more successful and people are living longer. Between 1960 and 2014, the global infant mortality rate dropped to one quarter of the 1960 rate. The average life expectancy also increased by more than 36 percent in the same time.³² The graph above looks something like a binary switch with the human presence on earth turning from virtually “off” to completely “on” primarily in the last 250 years. It is worth noting that the graph in figure 2-1 only looks at historical data. There is evidence that the population growth (as a rate) is decreasing with declined fertility rates in recent years. Steady growth numbers will also slim the realized rate over time, as a constant increase becomes a smaller percentage of the larger total in each year. The UN predicts that the growth rate will level out by 2100 however it should be noted that this prediction does not appear to consider potentially large innovations in health technology which could further increase life expectancy.

In addition to our great population boom, many of today’s theories hold that the growth rate of human technological power is exponential. A common theory, known as Moore’s law states computing capacity will double every 18 months. This has held approximately true for more than seventy years now since the first computer was invented. New theories propose that

³¹ Worldometers.info/world-population, Population past and present (2016)

³² World Bank Data - <http://data.worldbank.org/>

the historical eighteen-month window is likely shortening.³³ Serial entrepreneur, inventor, futurist and author of *The Singularity is Near*, Ray Kurzweil developed a theory called “the law of accelerating returns” which explains why technology and the human evolutionary processes in general progress in an exponential fashion.³⁴ A look at the recent history of human technology makes this theory relatively easy to defend. There is a logical premise that the more tools and resources humans create, the more technology we can and will build with those new tools. This snowballing momentum of tooling progress built on new technology is exactly the exponential effect Kurzweil and others before him have described. In his 1961 book, *Science since Babylon*, Derek Price calls this the “law of exponential increase”. As he explains it, “each [scientific] advance generates a new series of advances at a reasonably constant birth rate, so that the number of births is strictly proportional to the size of the population of discoveries at any given time.”³⁵

The ever-increasing pace of technology is responsible for a profound new human capability. Looking at the history of how humans have improved some of our most critical natural abilities provides many clear examples. Things that would have been essentially unimaginable just 200 years ago have become completely ubiquitous today. In physical mobility, we are now capable of interplanetary space travel. In a more common example on earth, we have achieved discounted international airfares comparable in price to a haircut in many popular cities. In communication, today’s real-time online video chatting between corners of the globe is vastly more sophisticated than communications mechanisms from previous generations. In the 1800s, hand-written notes may have had to sail the pacific, board steam trains, and ride horseback taking weeks or months to deliver even simple messages.

We looked at population growth and technology growth but there is another element growing faster than ever too – digital data. The term “big data” was first used in today’s context more than two decades ago now.³⁶ Today the estimated Internet population is roughly 40% of humanity (or 3.4 billion people) and this is easily the greatest source of so much big data. According to a November 2015 Ericsson Mobility Report, the number of mobile service

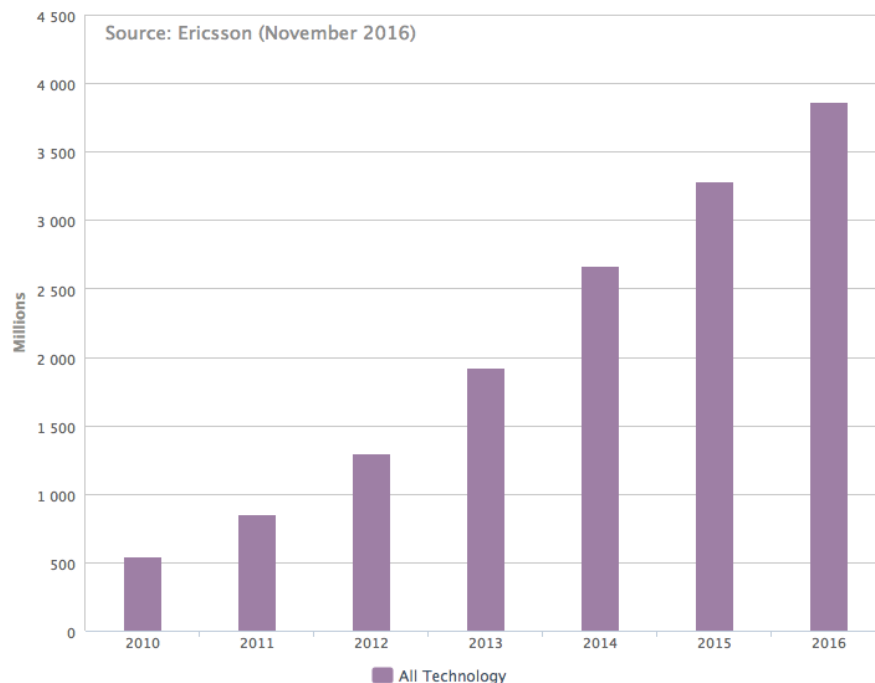
³³ Newlands, Murray and Read more. "The One Certainty About The Future Is The Pace Of Change Will Only Quicken". *Entrepreneur*. N.p., 2015. Web. 5 Dec. 2016 . <<https://www.entrepreneur.com/article/241255>>.

³⁴ Kurzweil, Ray - kurzweilai.net/the-law-of-accelerating-returns

³⁵ Press, Gill. *A Very Short History of Big Data*, Forbes.com, 9 May, 2013

³⁶ Press, Gill. *A Very Short History of Big Data*, Forbes.com, 9 May, 2013

subscriptions has already surpassed the global population.³⁷ With many individuals carrying multiple lines, the estimate suggests that actual unique mobile phone users make up closer to 4.9 billion people, or roughly two-thirds of the population. Also noted in the same November 2015 report, roughly 70% of the total subscriptions are for smartphone devices.³⁸



(Figure 1-4: global smartphone subscriptions)³⁹

While many theories on the future of our population propose we will naturally level out soon, as time passes, the pace of technology and data does not appear to slow for anyone or anything. In his 2014 address to the United Nations, 44th U.S. President Barack Obama proclaimed “Today, whether you live in downtown Manhattan or in my grandmother’s village more than 200 miles from Nairobi, you can hold in your hand more information than the world’s greatest libraries.”⁴⁰ With rapid globalization, human life as we know it is changing at an

³⁷ "Ericsson Mobility Report – ON THE PULSE OF THE NETWORKED SOCIETY.", *Ericsson.com* - Nov. 2015 *Mobility Report*. (page 2) N.p., 2015. Web. 12 Dec. 2016. <<https://www.ericsson.com/mobility-report>>.

³⁸ Ericsson Mobility Report (op. cit., page 2) Nov. 2015

³⁹ Ericsson.com. Ericsson Traffic Exploration Tool. Ericsson.com. N.p. 2016. Web. 3 Nov 2016 <<https://www.ericsson.com/TET/trafficView/loadBasicEditor.ericsson>>.

⁴⁰ Obama’s 2014 Whitehouse address to the U.N., <<http://www.whitehouse.gov/the-press-office/2014/09/24/remarks-president-obama-address-united-nations-general-assembly>>.

astounding speed and the world we once knew is becoming more connected constantly. As Bill Gates put it in a 2014 interview with the Atlantic, “I think the idea that innovation is slowing down is one of the stupidest things anybody ever said...Innovation is moving at a scarily fast pace”.⁴¹ Governments, which are tasked with enabling innovation and at the same time protecting citizens from harm, are finding it more and more difficult to keep up with technology. In the digital information age, individuals have relatively untethered access to instant global reach and more information online than was ever available before.

As new levels of knowledge and rapidly advancing digital capability enter the hands of our increasing population, there is a shift in power from governments to citizens (and the private organizations that employ citizens).⁴² A string of protests starting in 2010 referred to as the Arab Spring exemplify the new empowerment of people to share information, communicate and organize with or without the support of their government. With the power of social media communication, people demonstrated a new capability in the face of authority.⁴³ (It is important to note that the Internet is simply a tool and as such its forces will only be as good as the people that use them. Social media for example is also notoriously used by terrorist organizations to organize activity and recruit new members.⁴⁴ The ongoing shift of power from governments to people marks the increasing importance of education for individual citizens to manage themselves within the greater interest of society at large. As the old proverb goes, “with great power comes great responsibility”.⁴⁵ People are becoming more their own shepherds today, and the shapers of our current and changing environment.

Having covered some of the great changes we are experiencing, now we might ask, what are the associated challenges that humanity might face given these rapid changes? There are numerous ideas about where we are headed as a society, and as a species. As technology grows and the power of our species continues to expand, so too does the possibility that we could create our own fate. There are three essential types of manmade threats to humanity which we will

⁴¹ Freidman, Uri. *TheAtlantic.com/International*. 12 March, 2014.

⁴² Heimens, Jeremy and Henry Timms. "Understanding “New Power”". *Harvard Business Review*. N.p., 2014. Web. 6 Dec. 2016 . <<https://hbr.org/2014/12/understanding-new-power>>.

⁴³ Wasik, Bill. "#Riot: Self-Organized, Hyper-Networked Revolts—Coming To A City Near You". *WIRED*. N.p., 2011. Web. 11 Oct. 2016. <https://www.wired.com/2011/12/ff_riots/>.

⁴⁴ Wasik, Bill. *Wired.com*, 16 December, 2011 (www.wired.com/2011/12/ff_riots)

⁴⁵ Earliest notes from May 8, 1793 in a collection of the decrees made by the French National Convention <http://quoteinvestigator.com/2015/07/23/great-power/#note-11700-1>

explore: violence, famine, and technology.

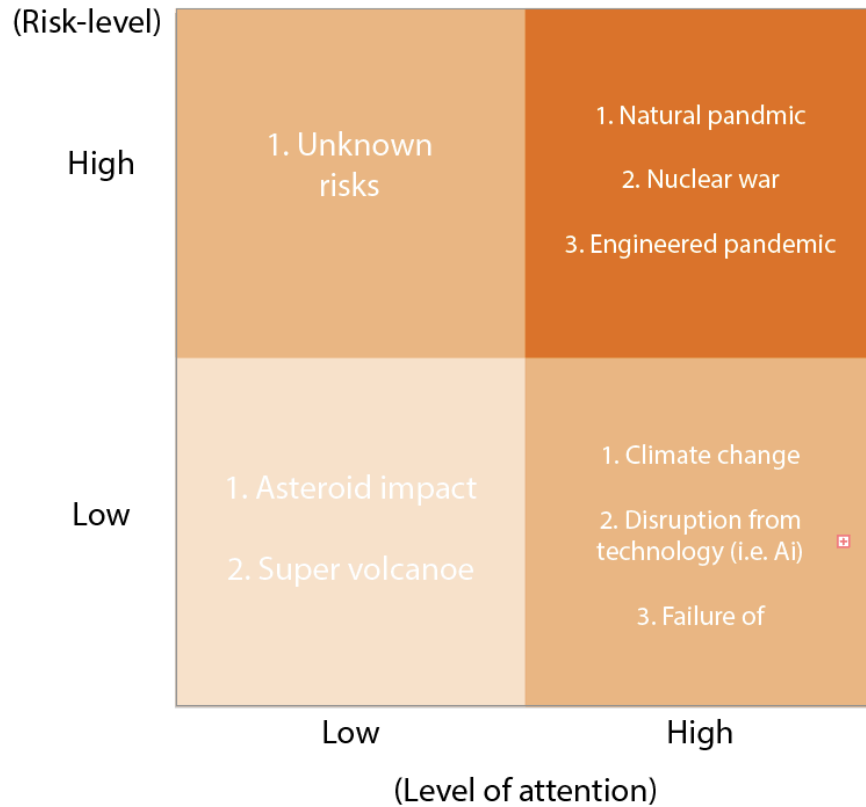
Stephen Hawking, a renowned theoretical physicist and a director of research at Cambridge University, is cited in a January 2016 article for his recent comments claiming that science and technology are responsible for most of the threats to our own species today.⁴⁶ More humans are alive than ever before and each of us is becoming more knowledgeable and powerful than our predecessors. In terms of our natural environment alone, it is difficult to comprehend the pace of human impact. By stacking population and technology trends side-by-side, it is evident that we have clearly become THE force to be reckoned with on earth. In December of 2015, Oxford Research fellow, Doctor Toby Ord presented a lecture outlining his long study on the most extensive research of extinction-level threats to humanity. The basis of his lecture is simply that natural potentials for human extinction (i.e. meteors or super volcanoes) are fewer and far less likely to occur than anthropogenic risks, like human violence, or pollution.⁴⁷

There are many issues that come from combining the rise of our population with the rapid adoption of quickly advancing technological power. As technology empowers greater levels of travel and communication, weapons of mass destruction have grown considerably and so too has pollution. Two primary types of threats to humanity result from the changes we are experiencing: new potential for violent destruction and new challenges in sustaining sufficient resources for our survival.⁴⁸

⁴⁶ Hawking, Stephen. "Black Holes Ain't As Black As They Are Painted, Professor Stephen Hawking: Black Holes, The Reith Lectures - BBC Radio 4". *BBC*. N.p., 2016. Web. 9 Dec. 2016 . <<http://www.bbc.co.uk/programmes/b06qjzv8>>.

⁴⁷ "Dr Toby Ord - Will We Cause Our Own Extinction? Natural Versus Anthropogenic Extinction Risks – CRASSH". *Crassh.cam.ac.uk*. N.p., 2015. Web. 17 Dec. 2016 . <<http://www.crassh.cam.ac.uk/gallery/audio/dr-toby-ord-will-we-cause-our-own-extinction-natural-versus-anthropogenic-e>>.

⁴⁸ Cotton-Barrat, Owen and Sebastian Farquhar. *<http://globalprioritiesproject.org/>*. N.p., 2016. Web. 14 Oct. 2016 . <<http://globalprioritiesproject.org/wp-content/uploads/2016/04/Report-2016-Executive-Summary-FINAL.pdf>>.(page 15)



(Figure 1-5. Mapped threats to humanity of the next five years)⁴⁹

Looking first at violence, the potential for human-inflicted violent destruction is specifically upheld by weapons of war and terrorism, both ballistic and biological. In a November 2016 article by the Wall Street Journal, titled *Obama seeks to Fortify Iran Nuclear Deal* there is mention of such threats to humanity. In the article, Iran is referred to as “The world’s largest organization of state sponsored international terrorism.” In his 2016 book “The Field of Fight,” retired general Flynn described correspondence between Iran and Al Qaida: “One letter to Bin Laden reveals that Al Qaida was working on chemical and biological weapons in Iran”⁵⁰ It still stands however that education in all ranks of society can create peaceful community and shared life purpose. As Dewey argued, education is a form of communication that creates mutual understanding and shared appreciation. By definition terrorism is the use of

⁴⁹ Image re-created from (GlobalChallengesFoundation, Global Catastrophic Risks 2016 report)

⁵⁰ Solomon, Jay and Carol E. Lee. "Obama Seeks To Fortify Iran Nuclear Deal". *WSJ*. N.p., 2016. Web. 17 Dec. 2016 . <<http://www.wsj.com/articles/obama-seeks-to-fortify-iran-nuclear-deal-1479692853>>.

violent acts to frighten the people in an area as a way of trying to achieve a political goal.⁵¹ Studies show that violent extremism is born of social situations that follow misfortune or violence.⁵² While a few well-educated leaders have participated in violent extremism, education is a well-known threat to the spread of terrorist ideologies⁵³. In the United Nations *Plan of Action to Prevent Violent Extremism*, UNESCO states its support for countries seeking to deliver education programs that build young people's resilience to violent extremist messaging and foster a positive sense of identity and belonging. This work is being undertaken within the conceptual framework of Global Citizenship Education.⁵⁴ The U.S. State Department and USAID *Supported Initiatives to Counter Violent Extremism* have educational initiatives woven throughout their goals. They note specific investments in STEM education and evidence-based critical thinking and values-oriented education.⁵⁵

Wars are historically the result of competing claims from stakeholders vying for limited resources.⁵⁶ When considering natural resource management, the most precious natural resource for humanity is clean water. It can be thought of as the blood of life itself. Without water the average human will only survive for about 3 or 4 days⁵⁷. Water also drives sanitation, which prevents greater levels of risk in the preservation of life. It is also worth noting that both access to clean water and the digital world rely heavily on energy resources. Water has been a centerpiece of UNESCO's initiatives for more than 50 years. In that time the population has more than doubled⁵⁸ and water consumption has tripled. Access to clean drinking water has been declared a human right by international law because it is fundamental to life. It was also estimated in 2010 that "about 1.5 million children under the age of five die each year and 443

⁵¹ The Definition Of Terrorism, Miriam Webster's Dictionary Online

⁵² *US Government - Strategy on Countering Violent Extremism*. N.p., 2016. Web. 4 Nov. 2016 . <<http://www.state.gov/documents/organization/257913.pdf>>.

⁵³ Beenish, Ahmed. "What's The Real Root Cause Of Terrorism". ThinkProgress. N.p., 2015. Web. Also

"United Nations Official Document". *Un.org*. Web. 24 Dec. 2015 . <http://www.un.org/en/ga/search/view_doc.asp?symbol=A/70/674>.

⁵⁴ "UNESCO". *UNESCO*. N.p., 2016. Web. 11 Dec. 2016 . <<http://en.unesco.org/>>.

⁵⁵ Obama's 2014 Whitehouse address to the U.N., <<http://www.whitehouse.gov/the-press-office/2014/09/24/remarks-president-obama-address-united-nations-general-assembly>>.

⁵⁶ How Resource Wealth Fuels War". *The National Interest*. N.p., 2015. Web. 12 Dec. 2016 . <<http://nationalinterest.org/feature/the-role-natural-resource-wealth-armed-conflicts-12195>>.

⁵⁷ Spector, Dina. "Here's How Many Days A Person Can Survive Without Water". *Business Insider*. N.p., 2017. Web. 6 Oct. 2016 . <<http://www.businessinsider.com/how-many-days-can-you-survive-without-water-2014-5>>

⁵⁸ Worldometers.info/world-population, Population past and present (2016)

million school days are lost because of water- and sanitation-related diseases”.⁵⁹ Water stress can quickly complicate any other environmental or health issues we might face. “In 2011, 41 countries experienced water stress, an increase from 36 countries in 1998. Of those, 10 countries, on the Arabian Peninsula, in Central Asia and in Northern Africa, withdrew more than 100 per cent of their renewable fresh water resources”.⁶⁰ Tom Pankratz, a desalination consultant and editor of the “Water Desalination Report,” has stated that there are just four methods to managing our clean water supplies; recycle more, increase storage, use less, or find alternative sources⁶¹. Water desalination, the process of creating clean drinking water from ocean water by extracting the salt, is a good option for increasing water supplies however it is still currently more expensive than other existing sources. The high price is largely based on energy though, and energy prices have been dropping with new technology.

A third and separate concern for humanity focuses on software, and it is wielding increased fears in the computer science community. A theory called singularity is focused on a point in the evolution of technology where humans create artificial intelligence which becomes more intelligent than the whole of the human species.⁶² There is a general consensus that singularity would represent sweeping societal changes with new capabilities but the jury is divided on whether the impacts of singularity would be good or bad for our species. In an ideal scenario singularity will drive us quickly towards a utopian future. If bad, singularity could also lead us to a chaotic dystopian struggle and even the eventual decline of our existence.⁶³

By leveraging education, we can frame the manmade threats explored in this section (violence, famine and technology) as opportunities to be exploited. Teaching culture and empathy can reduce our risk of violence. Through essential STEM education we can continue to develop better access to and management of valuable resources. Through education in computer

⁵⁹ 1st ed. Paris, France: ©UNESCO, 2015. Web. 11 Dec. 2016 .
<<http://unesdoc.unesco.org/images/0023/002350/235002e.pdf>>.

⁶⁰ "UN News - General Assembly Declares Access To Clean Water And Sanitation Is A Human Right". *UN News Service Section*. N.p., 2010. Web. 11 Dec. 2016 .
<<http://www.un.org/apps/news/story.asp?NewsID=35456#.WBa0l-ErLdQ>>.

⁶¹ Benkowski, Brian. "Desalination Is An Expensive Energy Hog, But Improvements Are On The Way". *Public Radio International*. N.p., 2017. Web. 6 Jan. 2017 . <<http://www.pri.org/stories/2015-05-15/desalination-expensive-energy-hog-improvements-are-way>>

⁶² "Singularity - Definition Of Singularity In English | Oxford Dictionaries". *Oxford Dictionaries | English*. N.p., 2016. Web. 5 Dec. 2016 . <<http://en.oxforddictionaries.com/definition/singularity>>.

⁶³ Newton, Richard. "2033: Grave New World Or Brave New World? Two Visions Of Life And Society In 17 Years Time". *Richard-newton.com*. N.p., 2016. Web. 6 Aug. 2016 . <<http://www.richard-newton.com/2033-grave-new-world-or-brave-new-world-two-visions-of-life-and-society-in-17-years-time/>>.

science, we will maintain better awareness and increase our understanding of the artificial intelligence systems, which we will rely on more and more each day.

Plato is often cited for a roughly translated quotation that “courage is knowing what not to fear.” This is derived from his story “Laches or Courage” where a character named Nicias exclaims:

“courage is the knowledge of that which
inspires fear or confidence in war, or in anything.”

– Laches (of Plato) ⁶⁴

One interpretation of this quotation is that we often fear what we do not know. On the contrary, should we know something well, we should not fear it, for we will already understand how to navigate its dangers. A similar interpretation of Plato’s quote is that if one is truly confident in his or her abilities (with knowledge), (s)he will no longer fear the unknown, believing (with confidence and courage) that (s)he will easily out-manuever any obstacles that lie ahead, known or unknown.

By reviewing prominent psychology and philosophy, this section serves as an observation of how empowering learning can be. Education is a mind-altering source of courage with the power to unite people, and help humanity transcend violence. Further, it is a tool for survival of the individual and the organism.

⁶⁴ Benjamin Jowett. *The Dialogues Of Plato*. 1st ed. New York: Random House, 1937. PDF. < pdf <http://izt.ciens.ucv.ve/ecologia/Archivos/Filosofia-II/Plato--Laches.pdf>

CHAPTER 2 – HISTORY AND EVOLUTION



(Image 2-1 from the opening scene from 2001, a Space Odyssey)⁶⁵

Segment - 2.1 –Dawn of Man to 1900 A.D.

Technology is defined as “science or knowledge put into practical use to solve problems or invent useful tools.”⁶⁶ We may take it for granted but written language itself is a fantastic early form of communication technology. During the opening scenes of the famous Stanley Kubrick movie “2001 a Space Odyssey” (1968) we observe an interpretation of the earliest use of a tool by our ancestral ape relatives.⁶⁷ The first tool is depicted as a bone simply used for a weapon, but Kubrick does a fantastic job emphasizing the dramatic discovery of a new tool. The monkey is excited by the power of his discovery in a basic way that any human can relate to. When realizing that she can hit things using a bone as a tool, the ape immediately has a wild tantrum of swinging and bashing. Human-centered technology has come a long ways since prehistoric

⁶⁵ Kubrick, Stanley, *2001, a Space Odyssey*. Burbank, CA: Warner Home Video, 2001

⁶⁶ The Definition Of Technology, *YourDictionary.com*

⁶⁷ Kubrick, Stanley, Arthur C. Clarke, Keir Dullea, Gary Lockwood, William Sylvester, Dan Richter, Ray Lovejoy, Geoffrey Unsworth, Aram Khachaturi, György Ligeti, Johann Strauss, and Richard Strauss. *2001, a Space Odyssey*. Burbank, CA: Warner Home Video, 2001.

times. New technology and scientific methods help to define the evolution of human design. New technology allows for new capability and re-draws the boundaries to expand design possibilities. This chapter looks at how popular education systems have evolved over time. We will trace technology breakthroughs and how they have impacted education design. After some history, we will discuss more contemporary education designs from the current century - a period of enormous technological breakthroughs.

To consider a prehistoric form of learning Salman Khan in his 2012 book “The One World Schoolhouse,” says to think of a “monkey see, monkey do” relationship whereby learning happens through basic observation.⁶⁸ The emulation of successful methods (and circumventing observed failure) is a form of learning in practice. Education then is a more formal approach to learning; we can call it the planned approach or a designed system for learning. For his earliest example of education Khan points to animal parents that educate their offspring on survival skills. Lions typically train their cubs by playing games in tall grass. The lessons of these games teach cubs how to stalk and hunt for prey. If this lesson is not learned well by the cubs, they will not survive. This stalking simulation is designed to pass knowledge between generations about hunting skills essential to life.

From a design perspective, this form of knowledge sharing is ideal in that it is already a *gamified* activity and therefore very engaging. Active learning participation has been studied extensively and engagement can be difficult to quantify but is commonly recognized as an important learning trait. More on engagement later once we dive deeper into challenges and goals of educational design.

We can jump ahead to humans for our next example of learning with language based knowledge sharing. Language is a game-changer for any species. “The transfer of more complex information, ideas and concepts from one individual to another, or to a group, was the single most advantageous evolutionary adaptation for species preservation.”⁶⁹ With the development of verbal communication, humans adapted the apprenticeship model of teaching and learning. Before school systems evolved, parents would teach their trades to their children. At home, girls learned from their mothers to look after the younger children and maintain the household.

⁶⁸ Khan, Salman. *One World Schoolhouse, The: Education Reimagined* (page 65)

⁶⁹ Ryan, Ph.D. Donald. *History Of Writing*. *Historian.net*. N.p., 2016. Web. 8 Nov. 2016 .
<<http://www.historian.net/hxwrite.htm>>.

Apprenticeships can be an excellent tool of the “observe and emulate” learning format. In his book Khan’s also points out how even present-day doctoral residency programs are a form of apprenticeship, where prospective doctors study and learn under practicing professionals of the medical field.

Next we find written language is easily broken into three periods: pre-printing press, post-printing press, and digital. In still ancient times, yet well before the printing press, clever people would communicate through cave paintings, hieroglyphs, and other drawings. Between approximately 3000 and 3500 B.C., most historians agree that various writing systems developed in ancient civilizations of the world.⁷⁰ It is said that the invention of written language took place when a king’s messenger in ancient Mesopotamia had too many times to remember and he thought to write things down.⁷¹ Introducing an alphabet that went beyond numbers and symbols came next. Compared to the earlier drawings and graphics-based communications, writing with an alphabet advanced the complexity and accuracy of concepts which could be shared. Drawings, paintings, and pictography were not comparable to an alphabet as they could not convey names; grammatical inflections; many verbs; and deeper, more accurate meanings.⁷² The oldest known alphabets were developed in central Egypt by 2700 B.C.⁷³ Since, alphabets have allowed a new level of knowledge sharing. Prior to the printing press, monasteries and religious centers typically housed very rare manuscripts. Written language was a phenomenal breakthrough in that it allowed human knowledge to be stored, transported and carried beyond its source. Now one person could attain, recall and share even more knowledge than they actually possessed.

Before the printing press, written language was incredibly difficult to find. In his acclaimed 2015 book, author and doctor Eric Topol suggest we imagine the cost to hire probably the most educated person in a town to copy by hand, written documents that were already scarce.⁷⁴ Learning how to write was even more difficult and so, small bodies of written material

⁷⁰ "Writing". *Ancient History Encyclopedia*. N.p., 2016. Web. 12 Dec. 2016 . <<http://www.ancient.eu/writing/>>.

⁷¹ Mark, Joshua, Priscila Scoville, and Cristian Violatti. On Writing - *Ancient History Encyclopedia*. N.p., 2016. Web. 12 Nov 2016 . <<http://www.ancient.eu/writing/>>

⁷² Daniels, Peter T. and William Bright. (Page 3). *The World's Writing Systems*. Oxford University Press. Printed, 1996. <<http://website.education.wisc.edu/~art908/wp-content/uploads/2014/10/writng-systems.pdf>>

⁷³ Crabben, Jan. "Alphabet". *Ancient History Encyclopedia*. N.p., 2011. Web. 2 Dec. 2016 . <<http://www.ancient.eu/alphabet/>>.

⁷⁴ Topol, Eric, *The Patient Will See You Now*. Basic Books 2015-01-06, Jun 2015, Chapter 3 (Audio book)

were primarily shared orally through public readings. There were many school systems and traditions, different all over the world; however, in almost every case (beyond perhaps a year or two of schooling in Greece) schools were exclusively reserved for religious leaders, royalty, and the extremely wealthy. Alongside the apprenticeship model, enthusiastic and relatively educated orators became a type of teacher, however lessons were incredibly inconsistent. The material taught was entirely subject to a speaker's collection of knowledge and personal understanding of topics.⁷⁵

Then, in the late fifteenth-century an education revolution began with the invention of the Guttenberg printing press. It was the most precise and advanced printing press on earth at that time. Topol describes the epic impacts of the printing press. Chapter three of this book emphatically describes how Gutenberg's fifteenth-century design was the single most transformative invention in all of human civilization. Topol then reflects on similarities between the changes observed following the printing press and current changes resulting from the modern day smart phone.⁷⁶ Topol also describes how before the printing press, people relied primarily on their ability to listen. Since use of the printing press humans have transitioned into relying primarily on their eyes. Only the highly affluent, nobility and priests had access to books before the printing press and could read - just eight percent of Europeans. The common man was left to listen to readings. Fairly quickly after the printing press, the public became literate as book printing grew in terms of soaring numbers and plummeting cost exponentially enhanced access.⁷⁷

“...[before the printing press] was a scribal culture with manuscripts that were remarkably expensive about a florin or roughly \$200 per 5 pages; or on average about \$20,000 dollars per book. The cost of the same book [after the printing press] would be approximately \$70, about 300 times less. It would take three individuals a lifetime to hand scribe 300 manuscripts. By 1470 three men could print 300 books in three months. Within 50 years [after the printing press] there were more books made than previously in the history of mankind. By 1500, an estimated 1,000 printing presses had spread throughout Europe and had produced millions of books. The error rate [before the printing press] was quite high, due to relying on a scribe, and

⁷⁵ Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. Chapter 8 (Audio Book)

⁷⁶ Topol, Eric, *The Patient Will See You Now*. Basic Books 2015-01-06, Jun 2015, Chapter 3 (Audio book)

⁷⁷ Topol, Eric, *The Patient Will See You Now*. Basic Books 2015-01-06, Jun 2015, Chapter 3 at 02:35 (Audio book)

manuscripts suffered from erosion, corruption and loss of content.”⁷⁸

“I’ll explain why the parallels can be seen as striking. Without question APP (after printing press) was associated with an explosion of knowledge and too much information. So too is our era. Back in the 15th century, as Nate Silver summed up; the amount of information was increasing much more rapidly than our ability to differentiate the useful information from the mis- truths. Here in the 21st century we call that big data. With more data generated in the past two years than in the history of human kind. And an ever increasing proportion of that is derived from and is passing through mobile devices.”⁷⁹

Still into the 1600s books were relatively scarce, however starting in the 1700s, books had become common enough (thanks to the printing press) to power something similar to what we know today as the typical school system. Teachers could read books along with students in the classroom and support student understanding. Khan explains how today’s public education systems have been primarily built on the Prussian eighteenth-century school model. It was a truly revolutionary model for its time. One of the Prussian school model’s most innovative contributions was that it was among the earlier tax-supported public systems - a huge breakthrough for education at the time. The Prussian model also had its flaws. Being of human design, as we have evolved it too should naturally evolve with modern perspective and standards for education. Salman Khan explains this well in the following excerpt –

“It is a human construct. It evolved along a certain pathway. Other pathways were also possible. Parts of the system we now hold as sacred, for example the length of class period, or the number of years we assign to elementary or high school, are in fact arbitrary, even accidental. Things that are now considered orthodox were at various points regarded as controversial and radical.”⁸⁰

The old classroom model segmented learning into primary and secondary classes, dissected knowledge into silos of subject areas, and even established the prototypical length and format of a school day and school year that is most commonly used today.⁸¹ Today there is a new breed of educational systems emerging online in e-learning - one which puts the responsibility, and

⁷⁸ Topol, Eric, *The Patient Will See You Now*. Basic Books 2015-01-06, Jun 2015, Chapter 3 at 07:35 (Audio book)

⁷⁹ Topol, Eric, *The Patient Will See You Now*. Basic Books 2015-01-06, Jun 2015, Chapter 3 (Audio book)

⁸⁰ Khan, Salman, *The One World School House*. Chapter 8 at 02:35 (Audio book)

⁸¹ Khan, Salman, *The One World School House*. Chapter 8 page 62 (Print)

therefore the power of learning, back into the hands of the student. Internet-based tools have been breaking down cost and location barriers to open up vast new levels of learning and access to education. These tools are rapidly commanding attention from the set of users worldwide that care about their education and are fortunate enough to discover them.

As beneficial as some of the Prussian school concepts may be, the content and delivery of this strict system is not well suited for autonomous learning. Students under this system cannot follow their own path based on curiosity or personal interest in subject matter. All students in a class are locked into tight cohorts that move students at a single pace through a uniformly mandated learning path. Additionally, the location of classroom instruction is fixed so that students need to travel varying distances to get to a physical classroom. Teaching pace might be too fast or slow at any given time for the individuals, and different barriers exist between each student and the physical classroom. We will come back to these challenges of pace, location, and direction of learning later with an open mind for what may be better suited to students individually.

Without passing too much judgment on the Prussian model, it is worth noting that it was built in the eighteenth-century. Any element of the Prussian model would therefore be worth re-evaluating with an acute consideration for new technology and a more modern societal climate.

Segment - 2.2 – 1900s Epistemology

During the early- to mid-1800s, the Prussian school model was widely adopted as the global model of a standard classroom. It is still in most cases what we are familiar with today from a large majority of early twenty-first century schools. There have been nominal innovations in the tools used (i.e. computers, whiteboards, books and writing materials). However, many older elements of the Prussian schoolhouse model (teachers lecturing on a fixed curriculum to students behind desks in physical classrooms) remain relatively unchanged in most present day classrooms around the world⁸². Starting in the late 1800s, new pedagogical theories were beginning to circulate, with some even tested and deployed by the early 1900s. Below we examine the findings from thought leaders on learning of this period. We will acknowledge their

⁸² Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. 1st ed. New York: Hachette Book Group, 2012. Print. (chapter 8)

theories on student abilities, and learning.

In 1907, following more than a decade of research, the Italian developmental psychologist and epistemology scientist, doctor Maria Montessori (1870 to 1952) launched her own independent school model in a province of Italy. Her new pedagogy was founded on many educational theories that she developed for herself. Most notably, Montessori held that all children are capable of learning, and that a student's unique learning ability is in fact only a result of their school's ability (or inability) to teach them.⁸³ This is a modern view that is shared more commonly today. She also argued that unique students are intended to learn within their own space, pace and path, guided intrinsically by each student's unique interests. She put responsibility for student learning back on the school's ability to nurture each student's interests in progressive ways, in turn encouraging those students to take real interest in their learning. These theories lead to practices very different from the traditional Prussian model. Some key differences include multi-age classes, variable learning time blocks, and variable inter-connected subject matter.

Since its founding, true Montessori school modeling requires that all teachers pass specific testing which acknowledges their ability and preparedness to adhere to Montessori's original methods. Today, the Montessori.edu website summarizes the following twelve core principles⁸⁴ of her methods that can be found throughout Montessori's career and the body of her various research papers and works:

- **Three-hour work periods**
- **Multi-age grouping**
- **Work centers**
- **Teaching method - "Teach by teaching, not by correcting"**
- **Teaching Ratio - 1:1 and 1:30**
- **Basic lessons - "that match student readiness"**
- **Interwoven areas of study**
- **Class sizes of 30 to 35 students**
- **Learning styles are nurtured – all types**
- **Assessment without grades, or other forms of reward or punishment**

⁸³ "The Montessori Method." *Digital.library.upenn.edu*. N.p., 2016. Web. 11 Dec. 2016 .
<<http://digital.library.upenn.edu/women/montessori/method/method.html>>. (introduction)

⁸⁴ "Montessori FAQ's General, Teachers, Schools". *Montessori.edu*. N.p., 2016. Web. 12 Dec. 2016 .
<<http://www.montessori.edu/FAQ.html>>.

- Requirements for age 0-6
- Requirements for ages 6-18
- Character education

“In 1914 Montessori went to America. She was welcomed by Thomas Edison and an American Montessori Society was formed with Alexander Graham Bell as its president”.⁸⁵ Montessori was also later nominated two times for the Nobel Peace prize, in both 1949 and in 1950 for her research.⁸⁶

Salman Khan, founder of KhanAcademy.com (a free online learning resource with millions of daily users), earned acclaim as a “groundbreaking innovator” from Bill Gates. His 2012 book, “The One World School House,” which filmmaker George Lucas referred to as “[A] must read for all who are committed to improving education”, is an in-depth look at the evolution of schools and an extensive preview of how learning takes place. His entire narrative is consistent with the views of other education specialists, however there are two learning concepts worth highlighting specifically from his text: mastery learning, and associative learning. Khan does not take credit for developing either idea. He explains how the concept has much earlier roots in an affluent Chicago school district around the 1920s.⁸⁷ Mastery learning was “predicated on the belief that all students could learn if provided with conditions appropriate to their needs... Second, mastery learning structured curriculum not in terms of *time*, but in terms of certain target levels of comprehension and achievement.”⁸⁸ Rather than teaching a lesson in a fixed amount of time (and returning variable student comprehension levels), mastery learning sets a fixed minimum level on each lesson comprehension, using a flexible course time dependent on the student’s needs to reach that level. To explore what he calls *associative learning*, Khan draws on the neuroscience studies of Nobel Peace Prize winner (2000) Eric R. Kandel. In short, there are two types of memory, short-term and long-term. The process where short-term memory becomes

⁸⁵ "Dr. Maria Montessori Biography - Daily Montessori". *Daily Montessori*. N.p., 2016. Web. 11 Dec. 2016 .
<<http://www.dailymontessori.com/dr-maria-montessori/>>.

⁸⁶ "Montessori Education UK". *Montessori.org.uk*. N.p., 2016. Web. 7 Dec. 2016 .
<http://www.montessori.org.uk/magazine-and-jobs/library_and_study_resources/teacher-training-study-resources/topics/peace>.

⁸⁷ Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. 1st ed. New York: Hachette Book Group, 2012. Print. (chapter 4)

⁸⁸ Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. 1st ed. New York: Hachette Book Group, 2012. Print. (page 38)

long-term is called memory *consolidation*. While not much is well understood about consolidation, it is known that when a new memory can be systematically associated with already established memories, both memories are strengthened⁸⁹. For example, while a random list of words may be difficult to remember, poetry makes a set of words more memorable (by weaving word association, meaning, and sound throughout the string of words).

A concept better known as active learning is generally defined as any instructional method that engages students in the learning process.⁹⁰ The active learning experiences can be classified along a spectrum of engagement, from passive at one extreme to active at the other. Another way to recognize active learning instructions is that they may require input directly from the learner that is relevant to flow of the lesson. There are different levels of “active” which can be measured by the amount of student participation in an exercise. In active learning, students are required to make choices, input elements, and think to determine the inputs of their lesson. Active learning is consistent with a popular theory of cognitive development known as constructivism. “The basic tenet of constructivism is that students learn by doing rather than observing.” In a simple analogy, it is noted that to learn how to ride a bike, you must get on a bike - you can’t just read a book about bike riding and expect any real capability. Constructivism seems to be included in the approaches defined by many 1900s developmental psychologist and educators, however the term was born out of the work of a Swiss-French philosopher and developmental psychologist Jean Piaget (1896-1980) and further developed also in the research of John Dewey.⁹¹ Piaget drew on inspiration from the Immanuel Kant (1724-1806), whose prior work arrived at a concept now known as constructionism. Constructionism is based on the concept of mental constructs, which dates back to fifth century B.C. philosophers like Lao Tzu (creator of Taoism) and Buddha⁹². A key focus of constructionism is exploring how a human will learn from their specific environment. Piaget’s constructivism ultimately highlights the same

⁸⁹ Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. 1st ed. New York: Hachette Book Group, 2012. Print. (page 48)

⁹⁰ Prince, Michael. "Does Active Learning Work? A Review Of The Research". *Journal of Engineering Education* 93.3 (2004): 223-231. Web.

⁹¹ "Constructivism Basics". *Edwebproject.org*. N.p., 2016. Web. 11 Dec. 2016 .
<<http://www.edwebproject.org/constructivism.basics.html>>.

⁹² "Constructionism And Constructivism". *Changingminds.org*. N.p., 2016. Web. 12 Dec. 2016 .
<<http://changingminds.org/explanations/research/philosophies/constructionism.htm>>.

importance that Montessori placed on building an appropriate learning environment in the classroom but Piaget went into much more detail about the specific sequence of environmental lessons that typically shape adult thinking.

Jean Piaget referred to his research as a study of *genetic epistemology*, or how people think and the origins of such thinking. His research (during the late 1800s to early 1900s) focused on how children develop into adults and looked carefully at the differences between how various children and adults solved problems. This work also led him to a concept he termed *cognitive theory*. Piaget's cognitive theory suggests that what separates children from adults is not mental ability but rather a collection of mental models that can be accessed for reasoning with new information and problem solving. His studies classified standard stages of mental development based on the mental models that people typically acquire during and between those stages. The mental models have been likened to mental building blocks that ultimately structure the adult reasoning abilities or intelligence.

Piaget considered intellectual growth as a process of adaptation to one's environment, using the actions of assimilation and accommodation to transition from a stage of confusion to one of understanding. Piaget's cognitive theory holds that confusion is uncomfortable and so, when encountering confusion there is an innate desire to gain understanding.⁹³ This theory introduces the importance of participation and engagement (via adaptation and assimilation) that is required by a student for learning to take place. Only when a student's understanding is appropriately challenged will they engage with the motivation to learn. This emphasis on student engagement was also highlighted in Montessori's original pedagogy, which is still in use today. As described by Linda Nussbaumer in her book *Human Factors in the Built Environment*, the Montessori model is based on two components: "self-motivation and auto-education, in which a child will learn naturally if put in an environment containing the proper materials."⁹⁴ This goes back to the innate human desire to learn and goes even one step further to suggest that learning is actually impossible if it is not first desired.

In a typical Montessori classroom as described by the Canadian Council of Montessori Administrators (CCMA), "you would see objects in baskets, trays, or boxes arranged on a shelf

⁹³ McLeod, Saul. "Jean Piaget | Cognitive Theory | Simply Psychology". *Simplypsychology.org*. N.p., 2016. Web. 12 Dec. 2016 . <<http://www.simplypsychology.org/piaget.html>>.

⁹⁴ Nussbaumer, Linda L. *Human Factors In The Built Environment*. 1st ed. Bloomsbury Academic, 2013. Print. (page 353)

attractively. Each work contains a purposeful work that is designed to teach a specific concept.” The CCMA also sites that “Rather than dictating what a child should learn and when, we design the classroom or home to fit the needs of the child, rich experiences balanced by beauty and order.”⁹⁵ By neatly presenting appropriate objects of learning in an attractive way, the Montessori-designed classroom model aims to help students discover and explore their own interests, creating a design to learn. As simple as it may seem, this approach ensures that the student goes from “passively” accepting what they will learn to “actively” choosing a subject of study. Students who have no involvement in their educational programming can easily become passive observers. Students of passive learning material are only encouraged to go with the flow of lessons as they are presented. Students of passive teaching are not given an opportunity to engage with their pedagogy. If not engaged in an exercise, they are more likely to miss essential material or misunderstand a core concept. Worse even, students may come to think of their lacking comprehension as a matter of being incapable.⁹⁶

Auto-education is defined as “Self-education - specifically learning shaped by the pupil’s own inclinations or needs”.⁹⁷ In “The One World Schoolhouse,” Salman Khan also explores at length the importance of auto-education. Khan states simply, but emphatically, that “taking responsibility for education is education”.⁹⁸ Khan describes how students in a traditionally passive classroom setting are actually denied of the appropriate responsibility for their learning. Khan plays on the example of a typical lecture hall class, where the teacher’s role is primarily to speak and the student’s role is primarily to listen. In this scenario, the teacher takes responsibility for 100% of the class content and the student is left to try and absorb anything that the teacher might say. Students can complete their role in this interaction without ever being required to think about the subject matter for themselves. To look at this scenario further, if a student is only in a lecture to fulfill a course credit, and has not chosen to be in the lecture for the purpose of learning, they are less likely to absorb the lecturer’s lesson. In this example of a passive learning scenario, it becomes far too easy for a student to blame the teacher (or other limited course materials) for a lack of understanding of the course subject matter.

⁹⁵ "CCMA - Canadian Council Of Montessori Administrators". *Ccma.ca*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.ccma.ca/>>.

⁹⁶ Khan, Salman, *The One World School House*. 2012 (page 55)

⁹⁷ "Auto-Education - Definition Of Auto-Education In English | Oxford Dictionaries". *Oxford Dictionaries | English*. Web. 17 Dec. 2016 . <<https://en.oxforddictionaries.com/definition/us/auto-education>>.

⁹⁸ Khan, Salman, *The One World School House*. 2012 (page 42)

Seymour Papert, a mathematician, computer scientist, inventor, and philosopher of epistemology, also researched issues associated with a passive pedagogy. In his 1980 book “Mindstorms,” Papert highlights the importance of natural student engagement within Piaget’s transitional model⁹⁹. He also develops on Piaget’s work creating his own definition for what he calls *transitional objects*¹⁰⁰, an object of study that the student is entirely responsible for selecting. Transitional objects, as he describes throughout the first chapter of his book, refer to a model that can illustrate a system with both abstract and sensory meaning¹⁰¹. In “Mindstorms,” Papert explains his early childhood relationship with mechanical gear sets. Motor gears were his transitional object from the age of two and he recounts the impact he believes gears had on his ability to understand complex concepts in his own early development. Interestingly, he describes falling in love with nearly every aspect of gear mechanics, including the way they felt when twisted between his fingers. “I saw multiplication tables as gears, and my first brush with equations in two variables (e.g., $3x + 4y = 10$) immediately evoked the differential.”¹⁰² He also asserts that “anything is easy if you can assimilate it to your collection of [transitional object] models”.¹⁰³ In a later section we recount how Montessori also embraces this concept that a student’s learning ability is only limited by the available tools and environment. Papert maintained a special appreciation for the epistemological framework of computer programming as a transitional object. He notes how when programming a computer, one must think about creating logic and creating new ways to think. As much as he favors the benefits of computer science as an ultimate form of his transitional object, he maintains that for programming to be a true object model, there must be an organic relationship derived of the student’s curiosities and personal interest. In other words, one cannot force a transitional object onto a student and expect it to stick. Papert goes so far in his definitions as to suggest that even the Montessori’s pedagogy is too narrow in its presentation of learning model options for student to study.¹⁰⁴

So what solutions did Papert propose to resolve the problems of passivity in pedagogy? He suggests a self-guided project-based learning framework where students decide exactly how

⁹⁹ Papert, Seymour. *Mindstorms* (1980), page 39

¹⁰⁰ Papert, Seymour. *Mindstorms* (1980), page viii

¹⁰¹ Papert, Seymour. *Mindstorms* (1980), page vii

¹⁰² Papert, Seymour. *Mindstorms* (1980), page 1

¹⁰³ Papert, Seymour. *Mindstorms* (1980), page vii

¹⁰⁴ Papert, Seymour. *Mindstorms* (1980), page vii

they would like to program computers. These are ideals of course and don't represent a real pedagogical proposal. As Papert puts it, "In many schools today, the phrase 'computer-aided instruction' means making the computer teach the child. One might say the computer is being used to program the child. In my vision, the child programs the computer and, in doing so, both acquires a sense of mastery over a piece of the most modern and powerful technology and establishes an intimate contact with some of the deepest ideas from science, from mathematics, and from the art of intellectual model building."¹⁰⁵ Active learning will lead to a better understanding of a topic through stronger memorization of core concepts that contribute to a subject. A greater level of engagement throughout a learning exercise will enhance the experience in terms of student learning.

Papert was obsessed with learning through computers. He spent more than two decades developing his proposal for computers in the classroom and was brought on as co-founder of the acclaimed MIT Artificial Intelligence Lab.¹⁰⁶ He also inspired the Lego Mindstorms toy set (a programmable Lego kit), which was named after his 1980s book and has been in productions for more than 19 years as of this writing¹⁰⁷. His obsession went deeper than just computers, and was founded somewhat on the mathematics which drive them. Like many, his work regards math as a language of logic, and programming as the computer language meant to perform such logic. In his words, "The computer is the Proteus of machines. Its essence is its universality, its power to simulate."¹⁰⁸

To briefly recap this segment on 1900s epistemology, let's consider six core concepts of learning which have emerged throughout the work of leading learning experts. These include an unlimited learning capability, learner individuality, active learning, associative learning, learning models, and auto-education. This list is not the most exhaustive list from a psychology perspective, but rather a short list of developmental psychology concepts primarily explored since the Prussian classroom model became a global standard. It would not be so easy to

¹⁰⁵ Papert, Seymour. *Mindstorms* (1980), page 14

¹⁰⁶ Blikstein, Paulo. "Seymour Papert's Legacy: Thinking About Learning, And Learning About Thinking | Transformative Learning Technologies Lab". *Tltl.stanford.edu*. N.p., 2017. Web. 9 Jan. 2017 .
<<https://tltl.stanford.edu/content/seymour-papert-s-legacy-thinking-about-learning-and-learning-about-thinking>>.

¹⁰⁷ LEGO.Com. *History - Mindstorms | LEGO.Com. Lego.com*. N.p., 2016. Web. 7 Nov. 2016 .
<<https://www.lego.com/en-us/mindstorms/history>>

¹⁰⁸ Papert, Seymour. *Mindstorms* (1980), page viii

consider all of these features of human psychology in the development of an education system, but remaining aware of these human learning traits should help to grade or even inspire effective pedagogy creation by a modern set of standards.

Segment - 2.3 – Digital Tools

Looking from the 1990s through the early 2000s we see a new set of tools being created in the onset of the digital age. This section introduces a few of the emerging eLearning (or online learning) tools that we see gaining popularity in present times. In the digital age there are a number of new learning tools cropping up which also break away from the standard classroom model. Personal computers and the Internet have allowed new access to various private digital learning resources, available 24 hours a day from the comfort of one's home. Most of the new privately held Internet learning services began as paid offerings, however many more now issue free unlimited user accounts too. While these new innovative educational resources are gaining in popularity every day, they are still not yet the norm for most students at present. To observe some of the more popular tools available today we will look at a recent evolution of digital resources provided by companies like Rosetta Stone, Youtube, Khan Academy, Stanford University and Duolingo.

Starting with Rosetta Stone¹⁰⁹ in 1992, audio recordings of language instruction were coupled with images and packaged onto digital compact disc format for distribution. This allowed owners of the CDROM package to access lectures and course materials at any time, in any place with the appropriate computer. This is a fantastic upgrade to accessibility, however students would still be required to purchase the physical CDROM somewhere and physically take it (or a copy of the software) with them for use. Next, the social video platform Youtube began inviting its visitors from any location to share and access custom video & audio content. Youtube has been a pioneer in what's called "user generated content" allowing anyone, but most importantly teachers or other subject experts, to record and share their own materials. One such person to begin uploading and sharing instructional video content with users in 2003, was Salman Khan. He explains in his book that his intention with Youtube was originally simply to share recorded math tutorials with his niece who lived in a different time zone. By making his tutorials public on the social platform, he unexpectedly and unintentionally gained thousands of

¹⁰⁹ Rosetta Stone®". *Rosettastone.eu*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.rosettastone.eu/about>>.

video views from random Youtube users seeking tutorials just like those he was creating. More than just views, given the social nature of the Youtube video platform, Khan also received much praise from thankful viewers in the form of user comments about the helpfulness of his videos. Today Khan Academy, a full-blown education platform, still based on the Youtube instructional video format, but has expanded to include interactive games with badges and points, quizzes and test, and much more. As of March 2015, the platform also includes a wide range of topics for learning including things like computer science, biology, and even art history. Khan Academy is a non-profit organization whose mission is “to provide a free, world-class education for anyone, anywhere.” On the About page of their website just below the mission, they describe the platform as “A personalized learning resource for all ages” and also note millions of daily users at this point. This additional service description highlights Khan’s appreciation for the pedagogical thinking of many before him (Montessori and Papert) who have suggested that learning is both unique to the individual (in terms of both pace and path) and not limited to any age.

Another innovative instructional model known as the MOOC¹¹⁰ (short for massive open online course) was first developed in Canada in 2008, and gained great popularity particularly within well renowned American universities like Stanford and Harvard who have insisted on being part of the innovation. MOOCs are in most cases quite similar to services like Khan Academy however they also often allow for genuine student-to-student and student-teacher interaction. MOOCs typically involve students watching video lectures, following along with any course materials (like outside software or texts), uploading completed assignments and taking online test and quiz materials. MOOCs typically involve a greater level of course community engagement and are also associated with a longer format of more in-depth segments than you find on Khan Academy. It’s not yet evident that either format is necessarily better or worse.

One last example of an emerging eLearning tool that has also recently reached mass adoption is Duolingo. Duolingo is a language teaching software launched in November 2010 whose “ultimate goal is to give everyone access to a private tutor experience through technology”.¹¹¹ The software utilizes a well-designed thoroughly gamified course structure. In

¹¹⁰ Marques, Juliana. "A Short History Of Moocs And Distance Learning" *Moocnewsandreviews.com*. 2013

¹¹¹ *Duolingo.com*. N.p., 2016. Web. 11 Dec. 2016 . <<https://www.duolingo.com/info>>.

2012 Duolingo conducted an effectiveness study, which followed beginner Spanish language learners and concluded that an average practice time of just thirty-four hours using the fun quiz-like challenges is equivalent instruction to a first semester university Spanish language course.¹¹² Duolingo was originally developed as a teacher and student project at Carnegie Mellon University. Co-founder Luis Von Ahn was also the inventor of Captcha and in both projects has utilized a rather genius method of pooling user quiz results to generate real-world monetary value. In the example of Captcha, which stands for “Completely Automated Public Turing test to tell Computers and Humans Apart,” when Internet users are presented with this test, they are actually digitizing legitimately un-readable text documents. In just twenty percent of the millions of instances where people take five seconds to prove that they are not computer programs, they are also helping to digitize old text documents. The first real-world project for reCAPTCHA was digitizing the entire New York Times’ archive which dates back to 1851.¹¹³ Now with Duolingo, as students answer language translation questions in their quest to learn new languages, they are actually also translating news articles from English into other languages for large digital publishers like BuzzFeed and CNN. The model is impressive and somewhat unique but so too is Duolingo’s user growth. In just six years since its launch, Duolingo now boasts more than 120 million users. It is also one of the premier examples of an eLearning system using machine learning methods to constantly improve itself.

¹¹² VESSELINOV, ROUMEN and JOHN GREGO. *Duolingo Effectiveness Study*

N.p., 2012. <<https://s3.amazonaws.com/duolingo-papers/other/vesselinov-grego.duolingo12.pdf>>.

¹¹³ Havel, John. "Recaptcha: The Genius Who's Tricking The World Into Doing His Work". *The Hustle*. N.p., 2015. Web. 11 Dec. 2016 . <<http://thehustle.co/the-genius-whos-tricking-the-world-into-doing-his-work-recaptcha>>.

SECTION 2

Modern design methods and Emerging Technology

CHAPTER 3 – DESIGN THINKING

A common mistake when first incorporating new technology into an existing system is using the new technology to simply re-build the same old system. A classroom example of this might be the use of iPads only to replace textbooks. While iPads are smaller, lighter, and capable of holding much more text than their paperback predecessors, this use alone is missing the point of having an interactive display on a smart device capable of connecting to the Internet. A similarly narrow approach is to only take the existing uses of a new technology and effectively place those uses into an existing system without adding to the core functionality. It is natural sometimes to think at first in the most uncreative terms of how to utilize new technology, however, innovative design is more likely to come from taking a step back and looking at what the goal of a (product or service) system is. Earlier in section one we laid out the literal objectives essential to well-designed education systems: audience, content appropriateness, content engagement (or activity), and accessibility. In addition, good design will appear uncluttered and simple to use. Let's now look deeper into the above objectives with a consideration for modern design thinking and usability. Below we will highlight three modern design methodologies and explore how these best practices can be applied to the designs of contemporary education tools to best reach their objectives.

Segment - 3.1 – Design Methodology

The first design methodology we will consider is call user-centered design (or human-centered design). Far too often, creators may begin their creative process by considering the technical limitations of their environment - available resources, hardware, software or other restrictive factors. Thinking about environmental limitations before looking at the human end-users can easily distract designers from their mission of creating a user-friendly product or service. Looking at outside factors first can prematurely restrict design thinking, creating a blind spot that overshadows the end goal to create something useful. Following a user-centered design approach is synonymous with applying usability evaluation. “Usability Evaluation focuses on how well users can learn and use a product to achieve their goals”.¹¹⁴ The first step of the modern design process is to understand the intended users and user-context for a product or service. Deciding the critical audience of any educational system will depend on that unique system’s outset and integrity. Then excellent design should aim to be as painless as possible for it’s users. Whether taking on a user-centered design approach (also called human-centered design), or a design-driven innovation approach, somewhere at the center of the task is the user. Users are more broadly known as the user-base, the audience, the user audience, or the target audience. As defined by usabilityfirst.gov, “User-Centered Design (UCD) is the process of designing a tool, such as a website’s or application’s user interface, from the perspective of how it will be understood and used by a human user.”¹¹⁵ User-centered design has been adopted by nearly all modern design thinkers in some form and is the underlying basis of many major design guidelines found universally. Design experts at every level have agreed that in many ways a user-centered design is what creating good design is all about. Many design resources are available today that resound this sentiment.

A second widely studied design philosophy now commonly referred to as “inclusive design” (or “universal design”) has been relatively undisputed, and is the subject of thousands of

¹¹⁴ "User-Centered Design Basics". Usability.gov. Web. 11 Dec. 2016 . <<https://www.usability.gov/what-and-why/user-centered-design.html>>.

¹¹⁵ User-Centered Design Basics". Usability.gov. Web. 11 Dec. 2016 .

scholarly articles.¹¹⁶ Inclusive design is a set of design principles with special consideration for underrepresented human challenges (both mental and physical). Inclusive design suggests that the more people a tool is able to serve, the more successful it will be. With inclusive design principles, not only will a learning tool serve more people, but it will also serve the individuals of the whole group better as well. Some successful institutions that have gone so far as defining their own inclusive design guidelines include large governments (usabilityfirst.gov and gov.uk), multinational corporations (Google/ IBM), famous design firms (Ideo, Frog, and Mule Design), and university entrepreneurship programs (NYU). With inclusive design methods it is imperative to consider serving underrepresented populations when defining the initial audience that a design will serve. This practice aims to ensure a broader reaching audience than is generally first anticipated. With over seven billion people on earth, good design needs to account for the vast varieties and levels of unique user abilities. Particularly in software, inclusive design has been rapidly increasing in importance because of the rapid global reach that new digital designs can achieve. With software applications like Google and Facebook now capable pushing design updates to billions of people, there is a broader range of unique user abilities (and disabilities) to consider.



(Figure 3-1 Curb-cut) extended

One of the earliest wide-spread case studies on inclusive design focuses on the creation of

¹¹⁶ "Inclusive - Microsoft Design". *Microsoft.com*. N.p., 2016. Web. 11 Dec. 2016 .
<<https://www.microsoft.com/en-us/design/practice/howwemake-section>>.

“curb cuts” or sidewalk ramps, which allow for better access to raised walk ways for handicapped veterans. In the mid 1940s, the city of Kalamazoo, Michigan installed curb cuts in a pilot program to aid employment of disabled veterans.¹¹⁷ As nearly all street-side walkways are raised ten to twenty centimeters above the street, before the pilot program, many stores, factories, and public facilities were not easily accessible to numerous veterans, elderly persons, and other physically handicapped people with walking impairments. This was especially true Kalamazoo, where there was an unusually large population of disabled veterans following World War II. After their installation, in a very short time, the newly constructed curb cuts proved to help various people, not just the disabled veterans whose complaints first sparked the curb cuts test program. Now everyone including mothers with strollers, delivery workers with large shipments, less agile elderly folks and various other visitors, could benefit from the increased ease of sidewalk accessibility due to sidewalk ramps.¹¹⁸ With inclusive design thinking, it can be argued that the original ninety-degree curb is of poor design that limits its use or restricts its own capability to serve the public. As stated by the United Kingdom Design Council in its 2006 design guidelines documentation titled, *The Principles of Inclusive Design*, “Inclusive design aims to remove the barriers that create undue effort and separation” within a product or service experience.¹¹⁹ The inclusive design approach can be used as a method for reducing unnecessary user exclusion. The less intuitive benefit of inclusive design is that, by removing barriers to outlying user groups, new designs can also improve the general usability for all members of an audience.¹²⁰ Inclusive design does not only work with physical disability like the architectural application noted by the Kalamazoo pilot project. The approach is translated seamlessly into software application design and all other types of product or service oriented design. In fact, in today’s technological landscape, where a single new software application is capable of reaching billions of individuals, inclusive design thinking is becoming more important to the usability of

¹¹⁷ Brown, Steven. "The Curb Ramps Of Kalamazoo: Discovering Our Unrecorded History | Independent Living Institute". *Independentliving.org*. N.p., 1999. Web. 11 Dec. 2016 . <<http://www.independentliving.org/docs3/brown99a.html>>.

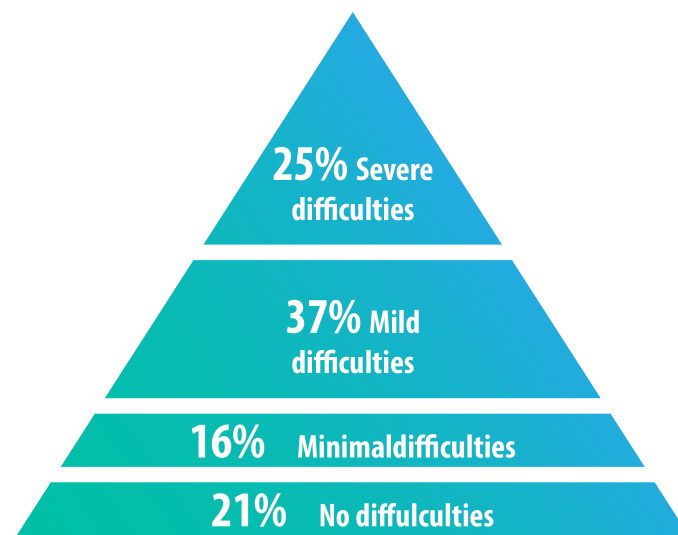
¹¹⁸ Loc. Cit. Brown, Steven. "The Curb Ramps Of Kalamazoo: Discovering Our Unrecorded History

¹¹⁹ "The Principles Of Inclusive Design: They Include You". *Design Council, London*. N.p., 2006. Web. 11 Dec. 2016 . <<http://www.designcouncil.org.uk/sites/default/files/asset/document/the-principles-of-inclusive-design.pdf>>.

¹²⁰ Loc. Cit. "The Principles Of Inclusive Design: They Include You". *Design Council, London*.

any broad reaching design.¹²¹

In 2003 Microsoft commissioned a usability research study conducted by Forester Research, Inc to test and quantify the difficulties adult computer users have where visual, hearing, dexterity, cognitive, and speech assistance would be helpful. The report concluded that for the average user of a computer-based service, the majority of people encounter some level of difficulty which could be alleviated by assistive services (including both accessibility options built into products as well as specialty hardware) that help individuals interact with a computer better. Below, figure 3-2 from the Microsoft study describes and highlights the findings which support the inclusive design approach which is more sensitive to the outlying variety of human capabilities.



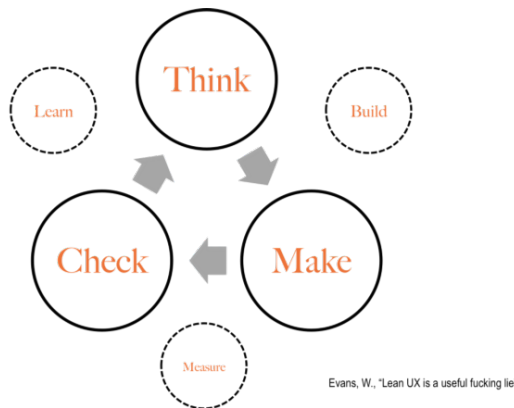
(figure 3-2 Agile design method)¹²²

A third modern design methodology to explore, is the increasingly data-driven design or test-driven development (TDD) approach adapted from Six Sigma, lean manufacturing, extreme programming (XP) and other business management systems of the past century. In the late

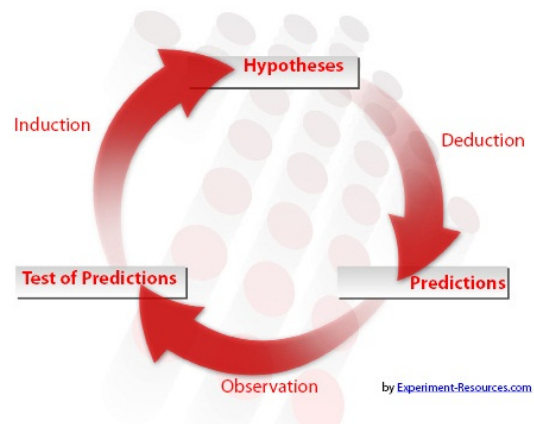
¹²¹ Pattison, Matthew and Alex Stedmon. "Inclusive Design And Human Factors: Designing Mobile Phones For Older Users". N.p., 2006. Web. 11 Dec. 2016 .
<<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.446.5036&rep=rep1&type=pdf>>.

¹²² Inclusive Design Toolkit. *What Is Inclusive Design?*. InclusiveDesignToolkit.com. N.p., 2016. Web. 7 Nov 2016 .
<<http://www.inclusivedesign toolkit.com/betterdesign2/whatis/whatis.html>>.

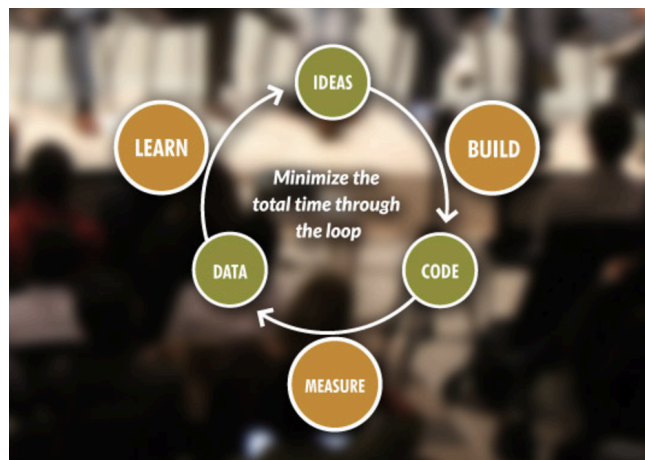
1990s, small start-ups and teams at large companies began to adopt well-proven iterative design frameworks.¹²³ Known as Lean, Scrum, Agile, and XP, these design methodologies are not entirely new practices; however, new levels of exposure and refinement in the 1990s helped them gain popularity as important modern design practices.¹²⁴



(figure 3-3: Agile design method)¹²⁵



(figure 3-4: The scientific method)¹²⁶



(figure 3-5: Lean design method)¹²⁷

¹²³ "History Of Lean Design® | Munro & Associates, Inc.". *Leandesign.com*. Web. Dec. 2016 .
<<http://leandesign.com/history-of-lean-design/>>.

¹²⁴ "What Is Agile? (10 Key Principles Of Agile) | All About Agile". *Allaboutagile.com*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.allaboutagile.com/what-is-agile-10-key-principles/>>.

¹²⁵ Original image inspired by "History Of Lean Design® | Munro & Associates, Inc.". *Leandesign.com*. Web. Dec. 2016 . <<http://leandesign.com/history-of-lean-design/>>.

¹²⁶ "What Is The Scientific Method?". *Explorable.com*. N.p., 2016. Web. 11 Dec. 2016 .
<<https://explorable.com/what-is-the-scientific-method>>.

With recent popularity over the past ten years, organizations like Google, IBM, Facebook and more have published guides on these methods. The promise offered by these methods is so strong that many observers site ‘manifestos’ of the methods in an almost religious following. In basic terminology, these methodologies all rely on a basic closed loop of testing and learning. What is particularly interesting about these systems is how they imitate and rely heavily on the traditional scientific method.¹²⁸ A scientific method is defined as “a method of research in which a problem is identified, relevant data are gathered, a hypothesis is formulated from these data, and the hypothesis is empirically tested.”¹²⁹ Another description has it worded that “the scientific method uses some type of measurement to analyze results, feeding these findings back into theories of what we know about the world”.¹³⁰ Essential to each of these related methods is how the final step of testing can feed into the first step where relevant data is collected.

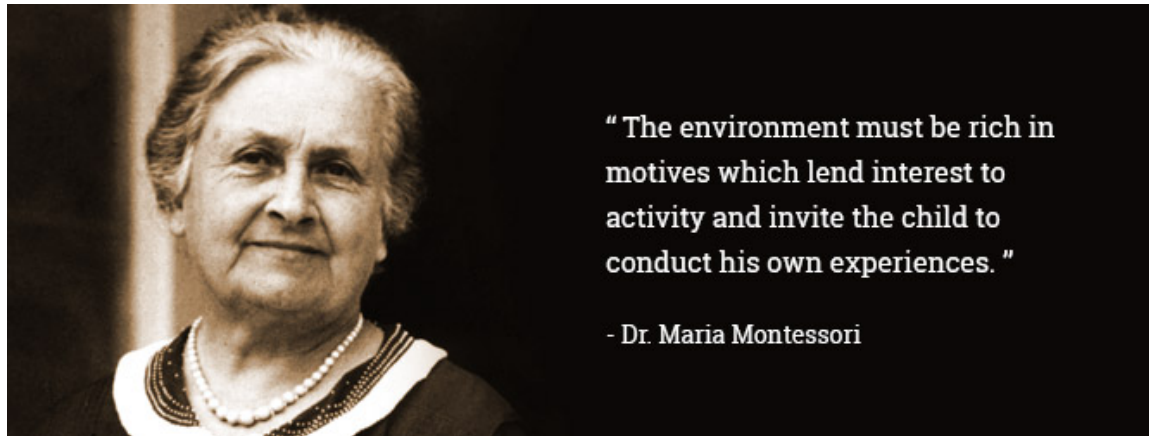
¹²⁷ Ries, Eric. *The Lean Startup | Methodology*. *Theleanstartup.com*. Web. 7 Dec. 2016 .
<<http://theleanstartup.com/principles>>.

¹²⁸ Mugridge, Rick. "Pages 1-3". *Department of Computer Science, University of Auckland, NZ*. N.p., 2003. Web. Dec. 2016 . <http://ftp://ftp.ufv.br/dpi/mestrado/TDD/Mugridge_TDD_Scientific_Method.pdf>.

¹²⁹ "The Definition Of Scientific Method". *Dictionary.com*. N.p., 2016. Web. Dec. 2016 .
<<http://www.dictionary.com/browse/scientific-method>>.

¹³⁰ "What Is The Scientific Method?". *Explorable.com*. N.p., 2016. Web. 11 Dec. 2016 .
<<https://explorable.com/what-is-the-scientific-method>>.

Segment - 3.2 – Case Study: Montessori Educational Design Methods



(Image 3-4 Maria Montessori)¹³¹

To recap, the previous section highlights three modern design approaches (user-center design, universal design, and lean design) which all can and should be applied to the design of any product or service. We now observe a case study very specific to the field of educational design. Maria Montessori fantastically applied all three methods over 100 years ago during the development of her own school model and as a result her learning system continues to be recognized for its value today.

With the support of her parents, a young Italian girl named Maria Montessori insisted on pursuing mathematics at a young age. She eventually even attended an all-boys school to do so, and after many years of persisting against the Italian system, in 1896 she became the very first female doctor in all of Italy. She began working at a psychiatric clinic where she fell in love with the children and thought deeply about ways to help them grow and develop into successful adults. She took careful note of the environment including toys and tools they were lacking and developed theories that education was in fact a better solution for developmental disabilities than most of the drugs being administered.¹³² After a few more years of observations from working with challenged youth, in 1899 she shared her theories at the Pedagogical Congress of Turin. As

¹³¹ "Preschool With Montessori Education System, Prayag - Bangalore, India". *Prayag.in*. Web. Dec. 2016 . <http://www.prayag.in/maria_montessori>.

¹³² "Dr. Maria Montessori Biography - Daily Montessori". *Daily Montessori*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.dailymontessori.com/dr-maria-montessori/>>.

a result of her presentation, Montessori was appointed as the directress of an orthophrenic clinic (school for the mentally ill) from 1899-1901. It was here that she would develop most of her essential theories, using the scientific methods she attained in medical school to validate her clinical observations.¹³³ After this experience Montessori left the clinic to conduct further studies in philosophical education and pedagogical pathology. Next, in 1906 she took on a daycare and teacher role for a very poor community in the new housing project of the district of San Lorenzo. She was given an assistant and was able to create her very own classroom model rooted in all of her theories and research of the past decade. The approach she took was highly experimental and her assistant had no experience in education, which she appreciated because it freed her of the traditional thinking behind standard school models. After just 12 months of this experimental experience (sometimes referred to as a living lab school) Montessori would open her first version of what is known today as the Montessori School model. Although Montessori's school model has gone in and out of style over the years, by 1925 her model was used at over 1,000 schools in North America¹³⁴ and today is currently followed at more than 4,000 schools across the United States.¹³⁵

A human-centered design perspective can be spotted in Montessori's early education theories (dating back to 1899) which proposed that education could be more effective than many of the drugs in use. In her theory Montessori's approach to problem solving (addressing learning disabilities in children) put the efforts of the people involved (the children) at the center of their own solution, rather than looking first at an unrelated technology like pharmaceuticals that might alter the brain to achieve an effect. In this way, Montessori simplified the problem at hand by bringing efforts closer to the center of a purely human challenge. Simplifying problems is a cornerstone of human-centered design. In keeping with user-centered design principles, Montessori also famously asserted that what had been deemed mental deficiencies were only in fact the results of a failed pedagogy.¹³⁶ Her statements were wildly controversial at the time,

¹³³ "Dr. Maria Montessori Biography - Daily Montessori". *Daily Montessori*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.dailymontessori.com/dr-maria-montessori/>>.

¹³⁴ "Montessori, Maria - Biographical". *Biography.com*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.biography.com/people/maria-montessori-9412528#educational-legacy>>.

¹³⁵ "Introduction To Montessori Method | American Montessori Society". *Amshq.org*. N.p., 2016. Web. 11 Dec. 2016 . <<https://amshq.org/Montessori-Education/Introduction-to-Montessori>>.

¹³⁶ Thayer-Bacon, Barbara. *InfactisPax*. N.p., 2003. Web. 4 Dec. 2016 . (page 3) <<http://www.infactispax.org/volume5dot3/ThayerBacon.pdf>>.

however her approach of placing responsibility on the pedagogical design itself (and not the student user) is particularly consistent with the suggested approach of modern user-centered design thinking. Another way to view the user-centered design perspective is to suggest that the customer (in this case student) is always right. If a product or service doesn't work for the customer, regardless of the reason, not serving the audience appropriately is a failure of the product's own design.

Still today the Montessori education model is cited for its explicit recognition that children as individuals will develop along unique paths and at various speeds in tune with the flow of their own unique interests. "The individual child would become the focus, the center of education; and the teacher, a director, who unobtrusively guided the child's own self-learning".¹³⁷ This underlying principle represents the mentality required by a universal design approach. More striking even is that Montessori studied education within the context of students who had specifically been diagnosed with learning disabilities. Because she was designing a system for, and practicing with, a developmentally challenged student audience, she was able to observe pain points in the educational model with more clarity. This type of user study epitomizes research under the inclusive design method. Setting out to problem solve for those with deemed disabilities is exactly what was also explored by the earlier curb-cut case study.

Finally, let's look at similarities between Montessori's approach and the well-proven agile design methods popularized in the 1990s. As a mathematician and medical student, Montessori was well versed in the scientific method. She began to work on what she called "scientific pedagogy." Scientific methods are the old code name for data-driven discovery. For a few years now in digital marketing and more commonly today in lean and agile (product or service) design, using data is a staple method for making sound decisions. Had previous education models only been the guesswork of their creators? It would appear so. In the later years of her work Montessori explains how up until her proposals for a new pedagogy, scientific methods had not been appropriately consulted in the design of popular education systems. She wrote specifically that

"To sum up the situation briefly, anthropology and psychology have never devoted themselves to the question of educating children in the schools, nor have the scientifically

¹³⁷ Montessori, Maria and Gerald Lee Gutek. *The Montessori Method*. 1st ed. Lanham, Md.: Rowman & Littlefield Publishers, 2004. Print. (page 3) < <http://www.arvindguptatoys.com/arvindgupta/montessori-new.pdf>>

trained teachers ever measured up to the standards of genuine scientists.”¹³⁸ Her scientific methods (a form of quantitative hypothesis and data-driven design) were arguably critical to the success of Montessori’s education models.

Well before modern design methods were so elaborately described and published, Maria Montessori managed to check a number of the most important boxes in the most modern design methodology. Her studies, passions, and consequent understanding of now popular developmental psychology served her well. Upon review of her research and her journey to creating what we know as the Montessori method, it is clear she benefited from an intuitive use of user-centered design thinking, inclusive design research, and even a relatively agile design method long before any of these frameworks were so formally defined.

¹³⁸ "The Montessori Method." *Digital.library.upenn.edu*. N.p., 2016. Web. 11 Dec. 2016 . <<http://digital.library.upenn.edu/women/montessori/method/method.html>>. (page 70)

CHAPTER 4 – EMERGING TECHNOLOGIES

In the 1910s & 1920s mechanics and engineers built machines increasingly capable of outperforming humans in jobs that traditionally required manual labor. This defines what is often referred to as the machine age.¹³⁹ Today, as software capabilities explode, ever-more complex decision-making is being run by computer programs that can replace the role of people in a similar fashion. The science of decision-making is quite literally just a complex human statistical analysis.¹⁴⁰ Once programmed, computer automated statistical analysis can be stunningly fast, perfectly accurate, and never needs to sleep. Combine our physical machines (e.g. cars or computers) with the latest software capabilities, and in the very near future many more human jobs, previously requiring a secondary and post-secondary education, will be replaced by software. Some examples are already taking place with simple customer support chat bots, and advancements in driverless vehicles. Given the trends, there is no doubt that software can soon even lead to improved educational programs, altering the traditional role of teachers in a big way. With simple math challenges, computer processing has been able to surpass human levels of accuracy and speed for decades (e.g. calculators). Digital formatting on a hard drive simply doesn't face the human challenges that our brain must overcome to remember and access sets of information accurately. As digital technology continues to advance, our tools become more robust but also more complex, and commanding them can quickly become more complex too. The designer's challenge is to reduce the complexity required to use a tool no matter how complex the tool's activities are. In this section we observe the latest trends in digital and communications technology, namely hardware (cell phones), their powerful software (mobile apps), and the mountains of new data that they both create. We conclude by looking at where some opportunities for improvement likely exist within digital [learning] tools.

¹³⁹ Walker, Patrick Gordon. "The Origins Of The Machine Age | History Today". *Historytoday.com*. N.p., 1966. Web. 15 Dec. 2016 . <<http://www.historytoday.com/patrick-gordon-walker/origins-machine-age>>.

¹⁴⁰ Butt, Fawad Iftikhar. "Statistical Thinking For Decision Making". *Aiu.edu*. Web. 12 Dec. 2016

Segment - 4.1 – Hardware, software, and big data

Already a few decades into the digital age, we now understand many of the transformative benefits of digital information formats (duplication, modification, location, physical size, and even the network effect) and the power of the Internet as the light-speed transportation highway of digital information. While the Internet infrastructure and the ‘ones’ and ‘zeros’ of digital formats stay mostly the same, three primary factors of digital technology (hardware, data, and software) continue to evolve and are therefore more difficult to keep pace with. Hardware development is cropping up in the form of smart devices. Software are those device’s operational programming scripts (or functional codes), and data are the additional information that can drive the software to create, record, manipulate, store and share information. These evolving devices, their increasingly sophisticated software and the mountains of new big data they are able to leverage now represent the fantastic new materials of contemporary digital design. As our computers and software continue to advance, they are able to access and manipulate larger and more robust sets of data. In just the past few years, the size of the data sets that a computer can access and process is rapidly catching up with human capability.¹⁴¹ Web 2.0 applications on today’s connected devices can harness what is called a *network effect*¹⁴² and are capable of working together over the internet in what is virtually a collectively powered hive mentality.

Let us first consider the hardware. When looking at the evolution of hardware today there is a focus on smartphones. The focus on smartphones is due to their combination of powerful processors and the rapid global penetration they continue to achieve. Powerful processors are able to support an increasing number of sensors power more sophisticated applications which access and crunch bigger data than before. In his 2011 book *Physics of the Future*, futurist and theoretical physicist Michio Kaku explains that;

“Today [in 2011], your cell phone has more computer power than all of NASA back in 1969 when it sent two astronauts to the moon. Video games, which consume enormous amounts of

¹⁴¹ Kurzweil, Ray. *The Singularity Is Near*. 1st ed. New York: Viking, 2005. Print.

¹⁴² Segaran, Toby. *Programming Collective Intelligence*. 1st ed. Sebastapol, Calif.: O'Reilly, 2007. Print.

computer power to simulate 3D situations, use more computer power than mainframe computers of the previous decade. The Sony PlayStation of today, which costs \$300, has the power of a military supercomputer of 1997, which cost millions of dollars.”¹⁴³

Our focus on smartphones is not to discount other smart devices (which can also access the Internet), however with 1.4 billion new smartphones sold in 2015 alone¹⁴⁴ their rapid adoption rate continues to outpace other standard smart devices. The inter-connective ability of all of these devices over the Internet allows them all to access and leverage new data in real-time. In the past decade, Internet growth is primarily supported by the growth of a thriving smartphone market. According to the 2015 Ericsson Mobility Report, for many people in the developing world, first contact with the Internet is now commonly happening via smartphones.¹⁴⁵

“Many consumers in developing markets first experience the Internet on a smartphone, usually due to limited access to fixed broadband. It took over five years to reach the first billion smartphone subscriptions, a milestone that was passed in 2012, and less than two years to reach the second billion. The four billion mark is expected to be reached by 2016.”¹⁴⁶

Since 2013 and 2014, a number of statistics have been released noting different variations of the same tipping point in which mobile Internet traffic has surpassed desktop traffic. It varies slightly depending on the activity (e.g. social media or email or online shopping), however Google confirmed that in May 2015, globally “more Google searches take place on mobile devices than on computers in 10 countries including the US and Japan”.¹⁴⁷ Comscore, a reputable Internet statistics company, also stated in April 2015 that there are more people who have “mobile-only” Internet access than there are with “desktop-only” Internet access. Mobile-only users accounted for 3.4 billion of the total data subscriptions considered.¹⁴⁸

¹⁴³ Kaku, Michio. "Page 118". *Future Physics*. N.p., 2011 <<http://www.brautech.co.za/info/FuturePhysics.pdf>>

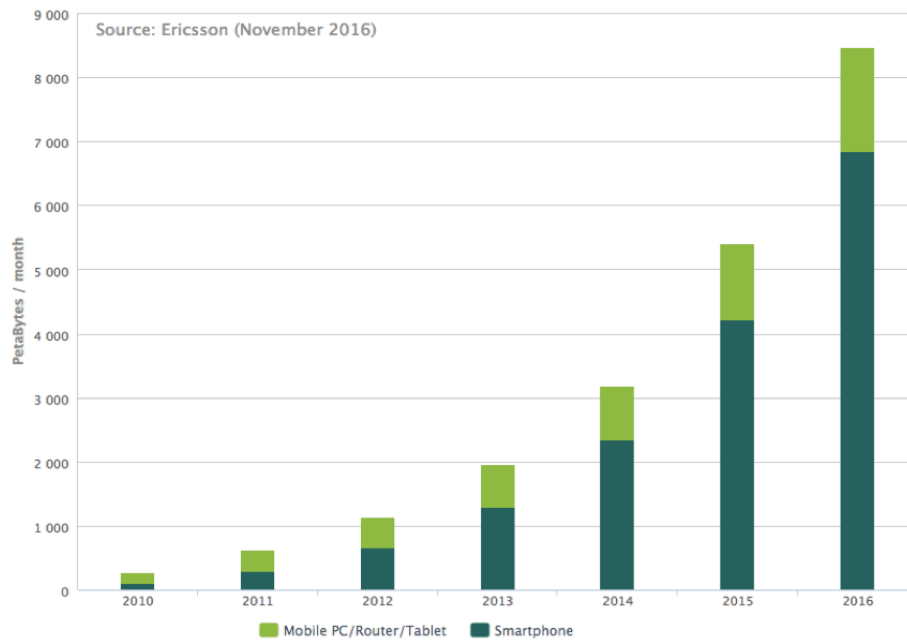
¹⁴⁴ Gartner Inc. Newsroom, *gartner.com* Egham, UK. Feb 2016. Web. 18 Feb. 2016

¹⁴⁵ Gartner Inc. Newsroom, *gartner.com* Egham, UK. Feb 2016. Web. 18 Feb. 2016

¹⁴⁶ Ericsson Mobility Report Nov. 2015 op. cit., page 7

¹⁴⁷ Dischler, Jerry. "Building For The Next Moment". *Inside AdWords*. 2015. Web. 17 Dec. 2016 . <<https://adwords.googleblog.com/2015/05/building-for-next-moment.html>>.

¹⁴⁸ Lella, Adam. "Number Of Mobile-Only Internet Users Now Exceeds Desktop-Only In The U.S.". *comScore, Inc.* Web. Dec. 2016 . <<https://www.comscore.com/fre/Insights/Blog/Number-of-Mobile-Only-Internet-Users-Now-Exceeds-Desktop-Only-in-the-U.S>>.



(Figure 4-1 growing Internet data usage – split by device)¹⁴⁹

The growth in smartphones (and consequent Internet access) is due to constantly falling hardware prices, particularly in the last few years. In 2014 Chinese technology producers began offering competitive smartphone models with a lower-range starting at just thirty US dollars.¹⁵⁰ Another important factor in the spread of Internet access is the significantly increasing amount of time that individuals spend on their devices (accessing the Internet). The Ericsson report notes that Internet data usage has increased across the board (on all devices), even in those regions where new Internet access growth is beginning to mature and slow down. Between 2014 and 2015, the report also notes a 40% increase in mobile data used per cellular data subscription.¹⁵¹ In its May 2016 public earnings call, Facebook founder, Mark Zuckerberg reported an increase in average daily use across its suite of software applications (including Facebook, Instagram, and

¹⁴⁹ Ericsson, *Traffic Exploration*. Ericsson.com. N.p., 2016. Web. 3 Dec. 2016 .
<<https://www.ericsson.com/TET/trafficView/loadBasicEditor.ericsson>>.

¹⁵⁰ Mirani, Leo. "\$30 Smartphones Are Here— And They'Re Getting Better Every Day". *Quartz*. N.p., 2014. Web. Dec. 2016 . <<http://qz.com/314285/30-smartphones-are-here-and-theyre-getting-better-every-day/>>.

¹⁵¹ Ericsson Mobility Report Nov. 2015 op. cit., page 2

WhatsApp) which jumped in 2016 to 50 minutes per day from 40 minutes per day in 2014.¹⁵²

This represents more information being accessed and created, whether it is text, photo, video or some other form of dynamic software.

Smartphones are also getting smarter as manufacturers compete with more robust sensors. Improved sensors can collect more types of data and larger, more detailed sets of data. A quintessential example is the smartphone camera with more megapixels offered every year. Today, cutting-edge smartphones are now offering two and three cameras per device. Some other common smartphone sensors (which can be constantly collecting more data) include the microphone, accelerometer, gyroscope, magnetometer, global positioning systems, light meter, barometric meter, screen touch meter (for manipulating touch screens), and LED sensors.¹⁵³ Many of these sensors are constantly active to help with the phone's functionality any time it is powered up. Functional examples include, listening for voice commands (Okay Google), brightening the screen in the dark, determining the proximity to your face for calls (to avoid accidentally pushing buttons), understanding if the phone is in your pocket and much more. These examples are functions often pre-programmed into a phone (by the device manufacturer) or installed by applications (which are granted access to the device sensors), and they do not represent the full capabilities of the sensors.

SOFTWARE - The sensors mentioned above are simply physical tools. The value of any software application is the output produced by using any of those sensors. In one example of this, the mobile application Vivino uses a phone's camera to scan and recognize wine bottles by brand mark, wine varietal and batch. The camera sensor is simply a hardware manipulated by Vivino software to create a new wine service. In fact, new creative manipulation of the sensors (via mobile applications) continues to extend the capabilities of all smartphones, even relatively older models. For example, an iPhone 4 device from 2012 (when equipped with a 2016 version of the mobile application for Google Maps) is robust of fantastic new features only developed in 2015 and 2016, such as off-line access to user locations on pre-cached maps. The various sensors in

¹⁵² Stewart, James. "Facebook Has 50 Minutes Of Your Time Each Day. It Wants More." *Nytimes.com*. N.p., 2016. Web. Dec. 2016 . <<http://www.nytimes.com/2016/05/06/business/facebook-bends-the-rules-of-audience-engagement-to-its-advantage.html>>.

¹⁵³ Caddy, Becca. "Here's How Your Phone Is Tracking You Right Now". *TechRadar*. N.p., 2016. Web. Dec. 2016 . <<http://www.techradar.com/news/phone-and-communications/mobile-phones/sensory-overload-how-your-smartphone-is-becoming-part-of-you-1210244>>.

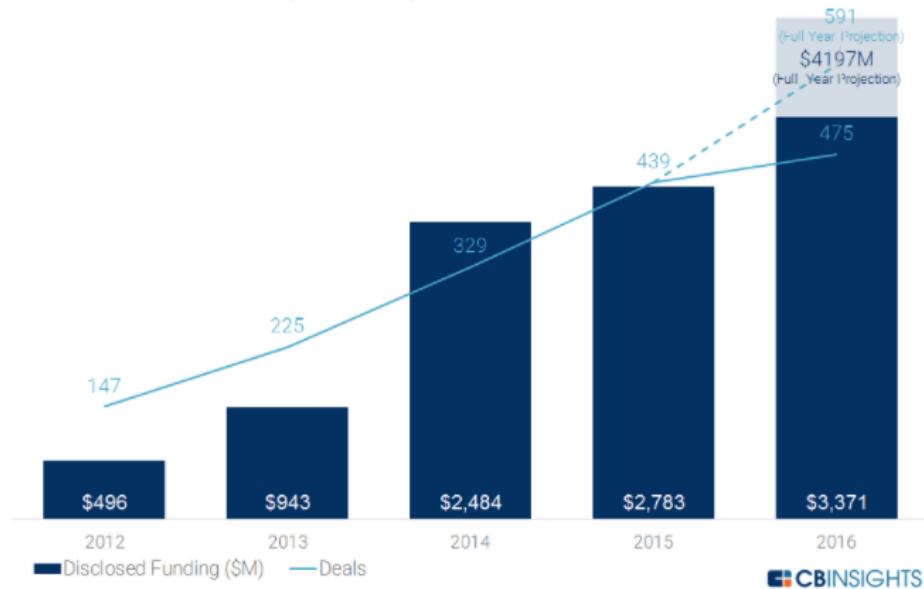
smartphones are for designers to consider and for the software engineers to exploit in developing new end-user capabilities. Many innovative application designs of recent are still constantly leveraging the same decades-old sensors in new and creative ways to get more and more utility out of smartphones. A nice example of a newer technology based on older hardware is facial recognition. Modern facial recognition software can be built today with fifteen-year-old hardware – a digital camera in this case. In addition to the old digital camera sensor, modern software programming techniques are what hold this technology together.

A particularly interesting area in software development is machine learning. There is frequent confusion and ambiguity surrounding when to use the term artificial intelligence (the computer simulation of organic or human intelligence), and when to use machine learning (software programs which are able to learn or improve themselves with new data). The reason for confusion is that there is often overlap. Ideal intelligence (artificial or human) should be capable of learning, and also machine learning is currently powering many examples of the most sophisticated artificial intelligence. Ramping in the past decade, machine-learning software has been disrupting many industries with its improvements for highly specific decision-making accuracy and speed. The promise that machine learning can create innovative artificial intelligence has attracted enormous attention recently. Virtually all of the largest global technology companies (Apple, Google, Facebook, Microsoft, IBM, Uber and more) are constantly increasing their investment in artificial intelligence (AI). Headlines over the past year have proposed that there is something of an arms race between the tech giants in the AI field.¹⁵⁴ In 2016 we also saw the formation of Open AI. Open AI's mission "to build safe AI, and ensure AI's benefits are as widely and evenly distributed as possible", has also been expressed as an effort to combat all of the outgrown corporate interests in the sector.¹⁵⁵ Popular computer scientist predictions suggest that there are revolutionary break-throughs coming soon, however

¹⁵⁴ Vance, More. "The Race To Buy The Human Brains Behind Deep Learning Machines". *Bloomberg.com*. N.p. Web. 10 Dec. 2016 . <<http://www.bloomberg.com/news/articles/2014-01-27/the-race-to-buy-the-human-brains-behind-deep-learning-machines>>.

¹⁵⁵ Metz, Cade. "Inside Openai, Elon Musk'S Wild Plan To Set Artificial Intelligence Free". *WIRED*. Web. Dec. 2016 . <<https://www.wired.com/2016/04/openai-elon-musk-sam-altman-plan-to-set-artificial-intelligence-free/>>.

it is difficult to nail down exactly what should be expected¹⁵⁶. At any rate, venture capital investment in artificial intelligence has increased almost 680% since 2012.



(Figure 4-3 Global Investments in Ai Technology)¹⁵⁷

Another way to view the core value of machine learning software is by recognizing its ability to distill large sets of big data into more meaningful information. Machine learning can help convert the base-levels of the data pyramid which Russell Ackoff explains (in segment 1.1) into higher levels achieving wisdom or intelligence (i.e. artificial intelligence).

¹⁵⁶ Newton, Richard. 2033: Grave New World Or Brave New World? Two Visions Of Life And Society In 17 Years Time. Richard-newton.com. N.p., 2017. Web. 6 Jan. 2017 . <<http://www.richard-newton.com/2033-grave-new-world-or-brave-new-world-two-visions-of-life-and-society-in-17-years-time/>>..

¹⁵⁷ Faggella, Daniel. "Venture Investments In Artificial Intelligence - Trends In 2016 And Beyond -". TechEmergence.com. N.p., 2016. Web. 15 Dec. 2016 . <<http://techemergence.com/investments-in-artificial-intelligence/>>.



(Figure 4-4 of source 123)

1. Data: symbols
2. Information: data that are processed to be useful; provides answers to “who”, “what”, “where”, and “when” questions
3. Knowledge: application of data and information; answers “how” questions
4. Understanding: appreciation of “why”
5. Wisdom: evaluated understanding

Another significant trend with software, relates to the fact that wars are generally fought over resources (as noted briefly in section 1.3, on page 20 of this text). In the last decade, as more of our resource control systems are based in digital, cyber warfare is gaining in its significance rapidly. In his 2016 book “Industries of the Future,” American technology policy expert Alec Ross explains this dependency on digital and the increasing importance of code. Chapter four of the book is aptly titled *The Weaponization of Code*. As a former cyber specialist of the Obama administration, Ross reports many interesting details, making the case that “The weaponization of code is the most significant development in warfare since the weaponization of fissile material”¹⁵⁸ (also known as nuclear weapons). In one example of the power possessed by hacking, Ross points to a 2013 instance where the Syrian Electronic Army purported from within the hacked Twitter account of the Associated Press (a respected multinational nonprofit news agency) the headline “Breaking: Two Explosions in the White House and Barack Obama is

¹⁵⁸ Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016). Chapter 4 – *The Weaponization of Code*

Injured.” This falsely published tweet triggered a value drop in the New York stock exchange of roughly \$136 billion in just two minutes. When it became clear the information was false, the stock market bounced back to its original levels. Ross notes how in 2011, United States Secretary of Defense Robert Gates declared “cyber” as a new domain of warfare alongside air, land, sea, and space. Ross also calls out a Chinese military unit known as People’s Liberation Army unit 61398, “thought to be the most advanced and best funded of the 20 cyber military units in China.” Ross mentions that “the cost of cyber attacks has eclipsed \$400 billion per year, a number that is larger than the gross domestic product of about 160 of the 191 countries of the world.”¹⁵⁹ One of the more interesting (however threatening) aspects of cyber warfare is the astonishingly low barrier to entry. Ross points out that while significant barriers (e.g. monetary and scientific) are in place to stifle offensive developments in nuclear and space, significant hacking offenses require particularly little resources. There is an entire host of complexities introduced by cyber attacks, from its completely ambiguous geographical orientation, to more fierce competition for talent due to lucrative investments of a booming private sector. To summarize much of his book, Ross makes it clear that software and cyber industries are producing nothing short of a revolution.

DATA – Reports among software companies with great interests in big data (including IBM) circulated in 2013-2014 that 90% of all the world’s data has been created in just the last two years^{160&161}. With the culmination of the four aforementioned hardware and user growth rates (more users, more activity, more sensors, and better sensors), we have experienced an enormous spurt in the amount of digital data created globally in a relatively short timespan. You don’t have to look further than a few of the largest Internet properties to understand how this is possible. Facebook, a company founded in 2004 has recently reported 1.79 billion monthly active users in November 2016. Further, 59% of those users are accessing the giant social media platform from “mobile-only” (smartphones only), an increase from 56.5% just 4 months earlier in 2016¹⁶². In

¹⁵⁹ Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016). Chapter 4 – *The Weaponization of Code*

¹⁶⁰ "IBM - What Is Big Data?". *Www-01.ibm.com*. N.p., 2016. Web. 12 Dec. 2016 . <<https://www-01.ibm.com/software/data/bigdata/what-is-big-data.html>>.

¹⁶¹ "Big Data, For Better Or Worse: 90% Of World's Data Generated Over Last Two Years". *ScienceDaily*. N.p., 2013. Web. 12 Dec. 2016 . <<https://www.sciencedaily.com/releases/2013/05/130522085217.htm>>.

¹⁶² "Facebook - Home". *Investor.fb.com*. N.p., 2016. Web. 15 Dec. 2016 . <<http://investor.fb.com>>.

addition to accessing more data, people are generating more data as more time is spent using the Internet or Internet enabled devices with applications like Facebook. We often don't think about how simply viewing an image on your phone can be an action that creates significant new data, but there are mountains of new data being built simply on tracking how users access information on the Internet (like text, photos, or websites). Let's imagine you are reading this paper on a digital device. You opened the document, possibly from a website or via an email. That's it! In this scenario, when the text opens there is almost certainly tracking software on the page (or in the text viewing application) tracing things like where you have arrived from (e.g. the email provider or website source), your device type, device model, its operating system (software brand & version), your location (post code, region, state, country), the date and time of your access to this text, how long you spend reading this text, any links you click through, and much more. In most cases this information is not *legally* allowed to be recorded and shared in personally identifiable ways. This means that the data from your visit should not be able to lead marketers who purchase the data to identify you on an individual basis. Typical of most internet marketing companies like Facebook and many others, Google mentions in its privacy terms that "To protect user privacy, Google policies mandate that no data be passed to Google that Google could use or recognize as personally identifiable information".¹⁶³ In this statement, Google requests that when installing Analytics software (their proprietary tracking technology) on a website or in an application, website administrators should avoid sharing things like a person's name or government ID number which can be tracked back to that specific person. By simply having your mobile smartphone turned on and connected, there is a good deal of data that gets saved to a data storage cloud somewhere overseas. Ross notes that "Private companies now collect and sell as many as 75,000 individual data points about the average American consumer, and that number is tiny in comparison to what's to come."¹⁶⁴ Examples of these various data points (often generated while smartphone users have applications open in the background) include:

- time / day / date / year

¹⁶³ "Best Practices To Avoid Sending Personally Identifiable Information (PII) - Analytics Help". Support.google.com. N.p., 2016. Web. 12 Dec. 2016 .
<<https://support.google.com/analytics/answer/6366371>>.

¹⁶⁴ Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016). Chapter 5 – *Data: The Raw Material of the Information Age*

- exact location (in wide a number of metrics)
- app usage status (frequencies / recencies / totals)
- scrolling / browsing / clicking data
- cookies in your browser or availability of other apps
- mobile device brand and model
- Internet browser version
- and much more

Other sensors like the camera, microphone, accelerometer, light sensor and more are further capable of capturing passive user-data. Technically, your connected smartphone could detect and report figures on how fast you are traveling and how quiet your commute is, from the passenger seat of your car.

So why is all of the big data being recorded, and how is it being used? Ross calls out that: “Big data is a catch-all phrase used to describe how these large amounts of data can now be used to understand, analyze, and forecast trends in real-time...There is a common misperception that the advances made possible by big data are simply a function of the amount of data gathered. In actuality, the growth in the amount of data without the ability to process it is not useful in and of itself.”¹⁶⁵

In many cases, the big data that a company collects may not be leveraged very well. As Ruben Sigala, the Chief Analytics Officer at Caesar’s Entertainment put it in an interview with McKinsey management consulting, “You have to start with the character of the organization”.¹⁶⁶ Each organization’s mission is unique and so each business case can be slightly different. Not everyone understands how to leverage big data, and particularly their organization’s unique data in a specific case for their core business objectives. Most people are beginning to understand that their organization’s data holds value, but until they understand the value, there is a difficult period of discerning which data is useful, and how to leverage it.¹⁶⁷

One meaningful factor in the recent rise of big data is the explosive use of mobile instant messaging applications. After voice communication dominated land-line telephones for most of

¹⁶⁵ Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016). Chapter 5 – *Data: The Raw Material of the Information Age*

¹⁶⁶ "How Companies Are Using Big Data And Analytics". *McKinsey & Company*. N.p., 2016. Web. 12 Dec. 2016 . <<http://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/how-companies-are-using-big-data-and-analytics>>.

¹⁶⁷ Olavsrud, Thor. "IT Wants (But Struggles) To Operationalize Big Data". *CIO*. N.p., 2016. Web. 11 Dec. 2016 . <<http://www.cio.com/article/3075423/it-strategy/it-wants-but-struggles-to-operationalize-big-data.html>>.

the twentieth century, mobile phones introduced text messaging. Text-based messaging gained huge ground and in 2014 it was confirmed to have outranked phone calls as the dominant form of communication with people under thirty-two years old.¹⁶⁸ It continues to grow still as instant messaging apps like Facebook, What's app, and hundreds more provide free domestic and international messaging services for those with Internet access. Of course those apps are collecting user-data and are leveraging an advertising business model, but they do not charge their users any money to use the services.

Segment - 4.2 – Interaction design, communication, and command

Having discussed recent trends in technology development, we are in a good position to consider opportunities to apply such technology to education. As mentioned at the start of this chapter, with more complex technology comes a [design] challenge to keep tools as simple and intuitive to use as possible. This means making them both un-cluttered (even while the features grow in number). The acclaimed German designer Dieter Rams expressed these challenges in his ‘Ten Commandments’ of good design.¹⁶⁹ Of his 10 design principles, numbers four and five speak to this design challenge best. Principle four follows that “Good design makes a product understandable. It clarifies the product’s structure. Better still, it can make the product clearly express its function by making use of the user’s intuition. At best, it is self-explanatory. Is unobtrusive.” Principle five states, “Good design is unobtrusive. Products fulfilling a purpose are like tools. They are neither decorative objects nor works of art. Their design should therefore be both neutral and restrained, to leave room for the user’s self-expression.”

Let’s observe three definitions that will help in understanding the relationship humans have with their technology:

- **Communicate:** “to transmit information, thought, or feeling so that it is satisfactorily

¹⁶⁸ Howe, Neil. *Why Millennials Are Texting More and Talking Less*. *Forbes.com*. N.p., 2015. Web. 15 Dec. 2016 . <<http://www.forbes.com/sites/neilhowe/2015/07/15/why-millennials-are-texting-more-and-talking-less/>>.

¹⁶⁹ "Good Design | About Vitsoe | Vitsoe". *Vitsoe.com*. N.p., 2016. Web. 12 Dec. 2016 . <<https://www.vitsoe.com/gb/about/good-design>>.

received or understood.”¹⁷⁰

- **Interaction:** “an occasion when two or more people or things communicate with or react to each other.”¹⁷¹

- **Command:** “The ability to use or control something.”¹⁷²

The terms above are listed in a specific order of general meaning. We can think of commands as a type of interaction, and interaction as a result of communication. The point being that to command our tools requires effectively communicating to them how we should like them to perform. How we are able to communicate with our tools will underpin how well we can understand and more importantly command them. The challenge with human-to-computer communication is simply that the core of computer programs typically only communicate in computer programming languages, yet only a very small portion of humans are skilled in those languages. Author Marijn Haverbeke describes his own 2012 computer programming book “Eloquent Javascript” as “a book about getting computers to do what you want them to do”.¹⁷³ It is of course, a book meant to help humans learn the second most common computer language, JavaScript. Further into the intro he explains the rising challenge of human to computer communication in this way:

“Computers are about as common as screwdrivers today, but they contain a lot more hidden complexity and thus are harder to operate and understand. To many, they remain alien, slightly threatening things. We’ve found two effective ways of bridging the communication gap between us, squishy biological organisms with a talent for social and spatial reasoning, and computers, unfeeling manipulators of meaningless data”.¹⁷⁴

On the other side of this communications challenge, it has generally been estimated that just 0.3% of the global population understands some form of computer programming language. As of

¹⁷⁰ “Communicate”. *Merriam-webster.com*. Web. 12 Dec. 2016 . <<https://www.merriam-webster.com/dictionary/communicate> >.

¹⁷¹ Dictionary, interaction. “Interaction Meaning In The Cambridge English Dictionary”. *Dictionary.cambridge.org*. N.p., 2016. Web. 15 Dec. 2016 . <<http://dictionary.cambridge.org/dictionary/english/interaction>>.

¹⁷² “Command - Definition Of Command In English | Oxford Dictionaries”. *Oxford Dictionaries | English*. Web. 12 Dec. 2016 . <<https://en.oxforddictionaries.com/definition/command> >.

¹⁷³ Haverbeke, Marijn. *Eloquent Javascript*. 1st ed. San Francisco: No Starch Press, 2011. Print. <pdf http://eloquentjavascript.net/Eloquent_JavaScript.pdf>

¹⁷⁴ Haverbeke, Marijn. *Eloquent Javascript*. 1st ed. San Francisco: No Starch Press, 2011. Print.

their August 2015 Developer Insights Report, IDC in conjunction with the Application Developers Alliance reports that:

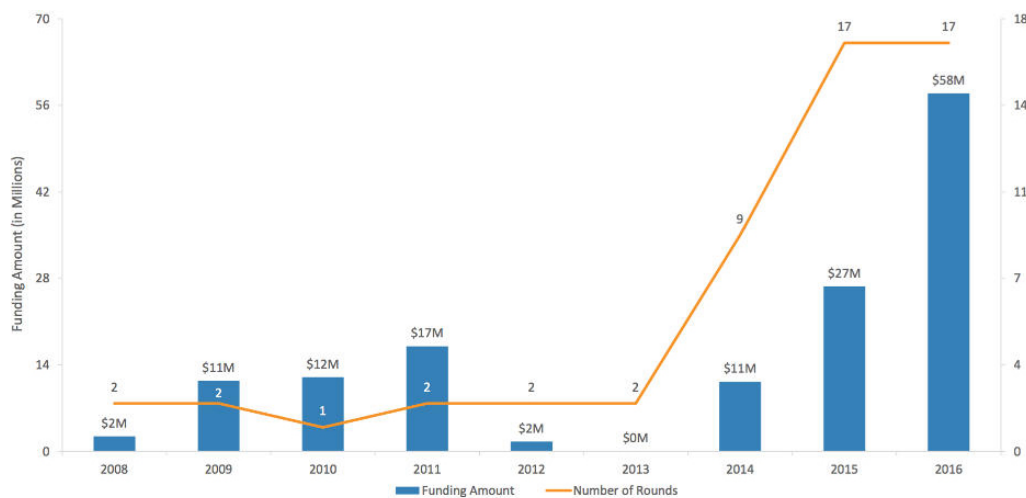
“IDC’s long-running tracking of worldwide developer populations pins the total number of developers at 18.5 million. This number is based on a bottom-up methodology of estimating developers according to labor and education statistics available for a number of countries”.¹⁷⁵

To consider how relatively few people can communicate in computer languages we may contrast those figures with roughly 20% of the human population which speaks Chinese (nearly two billion people globally). Given the ease with which computers can share their programs between each other, especially those connected over the Internet, it makes more sense to train computers to communicate in traditional human languages than it makes to try teaching all humans a variety of constantly evolving programming languages. This is where the science of natural language processing comes in. With the help of John Dewey (in chapter one) we established that education is actually a form of communication. In chapter two we observed the close link throughout history between communication technology and education tools. Circling back to that link, the impact of machine learning on current communication technology is noteworthy. In simple terms, natural language processing software is the use of algorithms that allow computers to process and understand human languages. The field is heavily based on machine learning and “covers areas such as sentence understanding, automatic question answering, machine translation, syntactic parsing and tagging, sentiment analysis, and models of text and visual scenes”.¹⁷⁶ These advanced language-processing algorithms are experienced by end-users in the form of translation programs and chat bots which can act as helpful agents like personal assistants and customer support representatives. All of these services are excellent examples of artificial intelligence and specifically an area of artificial intelligence that is quietly

¹⁷⁵ *Developer Insights Report*. 1st ed. Application Developer Alliance, 2015. Web. 15 Dec. 2016 .
<<https://static1.squarespace.com/static/53864718e4b07a1635424cdd/t/55de3d9ee4b07afa63281bf2/1440628126432/Apps+Alliance+Global+Developer+Insights+Report+2015.pdf>>.

¹⁷⁶ "The Stanford Natural Language Processing Group". *Nlp.stanford.edu*. N.p., 2016. Web. 15 Dec. 2016 .
<<http://nlp.stanford.edu/>>.

transforming software services as we know them. In a 2016 VentureBeat article¹⁷⁷, it is stated, that there are “Inflection points being reached in artificial intelligence and natural language processing that enable 90%+ accuracy in machine parsing and understanding spoken or typed requests.” In addition, many natural language-processing APIs have been made available, allowing more engineers to leverage the technology breakthroughs typically built in research silos. Some technology giants like Google lead the sharing, offering access to their “Powerful Text Analysis” tools.¹⁷⁸ These breakthroughs in language processing over the past few years have fueled a wave of investment in chat bots which promise to improve human and computer commands, communications, and interactions in general. Below we see a graph of the recent investment trend specifically for chat bots.



(Figure 4-6 Investment in chat bots)¹⁷⁹

The interesting take-away from this chart is that “Over \$140M has been invested in chat bots since 2010, with ~\$85M invested in 2015/2016.”¹⁸⁰

Of course when looking towards the future, predications are rarely exact, but by recognizing emerging communications trends we start to get a feel for what to expect in the

¹⁷⁷ Michael Quoc, Zipfworks. "The State Of Bots: 11 Examples Of Conversational Commerce In 2016". *VentureBeat*. N.p., 2016. Web. 15 Dec. 2016 . <<http://venturebeat.com/2016/06/16/the-state-of-bots-11-examples-of-conversational-commerce-in-2016/>>.

¹⁷⁸ "Cloud Natural Language API | Google. *Google Cloud Platform*. N.p., 2016. Web. 15 Dec. 2016 .

¹⁷⁹ "Tracxn Research - Chatbots Startup Landscape, June 2016". *Tracxn Blog*. N.p., 2016. Web

¹⁸⁰ "Tracxn Research - Chatbots Startup Landscape, June 2016". *Tracxn Blog*. N.p., 2016. Web

future of software interaction and also eLearning solutions. There is a wide range of ideas about how long it might take, but most signs suggest that we will be talking more fluidly with our software in the not too distant future.

CONCLUSION

To conclude let's review a few of the key topics explored in this text and then ask the question- "what should follow?" First we must consider that education is of human design and therefore should evolve with changing human needs. This means it is also bound to our human capability. We note that education is a form of communication and as such it has always evolved with our communication technology. The digital age continues to change everything, including human communication. We established a purpose for education: to prepare our species for adaptation to our changing environment. Given a population boom and the takeoffs of our exponential technological growth, the population, our pollution, and our own technologies clearly pose the greatest future challenges for humanity. Based on these leading challenges it is increasingly important we learn human empathy to coexist peacefully as a strong global community. It is also an increasing challenge for humans to maintain control of our rapidly advancing technology. If the human species is able to master these two domains, our potential is unlimited. Advanced computer knowledge and a deeper programming education may help our human species to safely navigate potential threats from our computers.

In our exploration of modern developmental psychology, we recognize six modern concepts related to human learning which can guide successful teaching tools of the future: uncapped learning capability, learner individuality, active learning, associative learning, learning models, and auto-education. It may be impossible to tackle all of these concepts in a single education tool, but these core concepts should inspire new and creative pedagogical designs. What is a student's relationship with his or her learning? How should students interact with their education? Ideally education tools should be dynamic and personalized, highly desired, deeply engaging and drive students to take maximum responsibility for their learning.

We also looked at keys to modern design thinking. These should help define processes which can guide greater design success (especially in education technology). These design methodologies include user-centered design, inclusive design, and any iterative data driven design method (e.g. lean design or the scientific method). Where do we begin the education design process? Who is the human audience we should study and serve? Following the

Montessori case study (in chapter three), I believe we will need to build education for different students who appear to hold learning disadvantage.

Thinking about the future of education design, I see two critical challenges. The first is a, design challenge and the second is a technical data challenge. However, both challenges could potentially be merged into a single challenge requiring design-informed technical executions. First, as our technologies have grown exponentially more complex and capable, the designer's challenge in the 21st century is increasingly to design intuitive technology interactions that remain clean and simple for their human end-users. Second, with the enormous data that digital world is creating, education tools must leverage user data to inform their design. Data scientists will need to develop software that learns from its interactions with students.

In chapter five of "Industries Of The Future," Ross describes translation software clocking billions of translations in a week, and the associated data rapidly outpacing the experience of career translators.¹⁸¹ On this note, online learning platforms like Khan academy will need to harness the great data that they generate from their millions of daily users. A company that does not aggressively leverage its user data is rapidly becoming analogous to a hunter wearing a blindfold. Even a heightened sense of hearing is unlikely to make one competitive in such a scenario.

There appear to be two approaches to maintaining control of our increasingly complex technologies. Either we learn to speak computer languages, or we teach computers to understand natural human languages better. In other words, maybe studying computer programming is not so important given recent progress in natural language processing. Some futurist are arguing that "Soon We Won't Program Computers. We'll Train Them Like Dogs".¹⁸² But let's also consider the benefits of epistemological thinking gained from studying computational programming. As Papert argued for years, computer programming provides an almost ideal transitional object for students to understand the rest of their world with.

¹⁸¹ Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016).

¹⁸² Tanz, Jason. *Soon We Won't Program Computers. We'll Train Them Like Dogs*. N.p., 2017. Web. 6 Jan. 2017 . <<https://www.wired.com/2016/05/the-end-of-code/>>

3 Designer Questions:

What areas of recent innovation in digital communication will shape the future of our educational platforms? Given the increasing use of eLearning platforms, and the advancements in machine-to-human communication, how as a designer, can I leverage the latest communication technologies, like chat bots (and other artificial intelligence), to help education software communicate more effectively with its human learners?

Considering the evolving challenges humanity faces (in terms of population growth and technology), how can modern education platforms teach people the empathy needed to live together peacefully? How, as a designer, can I utilize associative learning to direct the narrative that weaves both empathy and technology into a single lesson?

With all of the digital sensors and respective data points available today, we are opening up a lot of new potential capabilities – new (contextually informed), natural contact points to communicate with connected audiences in meaningful ways. Given the value that user-data can hold, how as a designer can I scientifically determine the most valuable user data in an educational context?

Bibliography:

ACASA. *Ackoff Center Weblog: Russell L. Ackoff, Management Consultant & Systems Thinker, 1919 -2009*.

Ackoffcenter.blogs.com 2009. Web. 7 Nov 2016 .

<http://ackoffcenter.blogs.com/ackoff_center_weblog/2009/10/russell-l-ackoff-management-consultant-systems-thinker-90.html>.

Ackoff, R. L. (1999) *Ackoff's Best*. New York: John Wiley & Sons, pp 170 – 172.

<<http://faculty.ung.edu/kmelton/Documents/DataWisdom.pdf>>.

Ahmed, Beenish. What'S The Real Root Cause Of Terrorism: Poverty Or Anger?. ThinkProgress. N.p., 2015. Web.

12 Nov 2016 . <<https://thinkprogress.org/whats-the-real-root-cause-of-terrorism-poverty-or-anger-d9d3f53324c5#.pzft3mmql>>.

Alphabet Inc. *Best Practices To Avoid Sending Personally Identifiable Information (PII) - Analytics Help*.

Support.google.com. N.p., 2016. Web. 12 Nov 2016 . <<https://support.google.com/analytics/answer/6366371>>.

American Montessori Society, *Introduction To Montessori Method*. Amshq.org. N.p., 2016. Web. 12 Nov 2016 .

<<https://amshq.org/Montessori-Education/Introduction-to-Montessori>>.

Bellinger, Gene, Durval Castro, and Anthony Mills. *Data, Information, Knowledge, & Wisdom*. Systems-

thinking.org. Web. 7 Nov 2016 . <<http://www.systems-thinking.org/dikw/dikw.htm>>.

Benkowski, Brian. *Desalination Is An Expensive Energy Hog, But Improvements Are On The Way*. Public Radio

International. N.p., 2017. Web. 6 Jan. 2017 . <<http://www.pri.org/stories/2015-05-15/desalination-expensive-energy-hog-improvements-are-way>>.

Blikstein, Paulo. "Seymour Papert'S Legacy: Thinking About Learning, And Learning About Thinking I

Transformative Learning Technologies Lab". Tltl.stanford.edu. N.p., 2017. Web. 9 Jan. 2017 .

<<https://tltl.stanford.edu/content/seymour-papert-s-legacy-thinking-about-learning-and-learning-about-thinking>>.

Brown, Steven. *The Curb Ramps Of Kalamazoo: Discovering Our Unrecorded History I Independent Living*

Institute. Independentliving.org. N.p., 1999. Web. 12 Nov 2016 .

<<http://www.independentliving.org/docs3/brown99a.html>>.

Buluswar, Murli: Campisi, Vince, and Gupta, Ash. *How Companies Are Using Big Data And Analytics*. McKinsey

& Company. N.p., 2016. Web. 12 Nov 2016 . <<http://www.mckinsey.com/business-functions/mckinsey-analytics/our-insights/how-companies-are-using-big-data-and-analytics>>.

Butt, Fawad Iftikhar. *Statistical Thinking For Decision Making*. Aiu.edu. Web. 12 Nov 2016

<<http://aiu.edu/publications/student/english/Statistical%20Thinking%20for%20Decision%20Making.html>>

Caddy, Becca. *Here's How Your Phone Is Tracking You Right Now*. TechRadar. N.p., 2016. Web. 17 Nov 2016 .

<<http://www.techradar.com/news/phone-and-communications/mobile-phones/sensory-overload-how-your-smartphone-is-becoming-part-of-you-1210244>>.

Cambridge Dictionary. *Meaning of Interaction In The Cambridge English Dictionary*. Dictionary.cambridge.org.

N.p., 2016. Web. 15 Nov 2016 . <<http://dictionary.cambridge.org/dictionary/english/interaction>>.

Canadian Council Of Montessori Administrators I CCMA. Ccma.ca. N.p., 2016. Web. 11 Nov 2016 .

<<http://www.ccma.ca/>>.

Carol E. Lee in Lima, and Jay Solomon in Washington. *Obama Seeks To Fortify Iran Nuclear Deal*. WSJ. N.p., 2016. Web. 20 Nov. 2016 . <<http://www.wsj.com/articles/obama-seeks-to-fortify-iran-nuclear-deal-1479692853>>.

Carvin, Andy. *Constructivism Basics*. Edwebproject.org. N.p., 2016. Web. 12 Nov 2016 . <<http://www.edwebproject.org/constructivism.basics.html>>.

Center For Universal Design NCSU. *About The Center - Ronald L. Mace*. Ncsu.edu. N.p., 2016. Web. 7 Nov 2016 . <https://www.ncsu.edu/ncsu/design/cud/about_us/usronmace.htm>.

Charter of the United Nations. *Chapter I: Purpose & Principle*. Un.org. N.p., 2016. Web. 12 Nov 2016 . <<http://www.un.org/en/sections/un-charter/chapter-i>>.

Cotton-Barrat, Owen and Sebastian Farquhar. *2016 Global Priorities Executive Summary Report*. <http://globalprioritiesproject.org/>. N.p., 2016. Web. 14 Oct. 2016 . <<http://globalprioritiesproject.org/wp-content/uploads/2016/04/Report-2016-Executive-Summary-FINAL.pdf>>.

Crabben, Jan van der. Ancient History Encyclopedia. *History of the alphabet*. Ancient.eu N.p., 2011. Web. 2 Nov 2016 . <<http://www.ancient.eu/alphabet/>>.

Curwin, Daniel. How Resource Wealth Fuels War. The National Interest. N.p., 2015. Web. 12 Nov 2016 . <<http://nationalinterest.org/feature/the-role-natural-resource-wealth-armed-conflicts-12195>>.

Daily Montessori, Dr. Maria Montessori Biography - Daily Montessori. N.p., 2016. Web. 12 Nov 2016 . <<http://www.dailymontessori.com/dr-maria-montessori/>>.

Daniels, Peter T and William Bright. *The World's Writing Systems. 1st ed.* New York: Oxford University Press, 1996. Print.

Darwin, Charles. *On The Origin Of Species By Means Of Natural Selection, Or The Preservation Of Favored Races In The Struggle For Life*. 1st ed. Down, Bromly, Kent, 1859. Print. <http://www.vliz.be/docs/Zeecijfers/Origin_of_Species.pdf>.

Dewey, John et al. *Democracy And Education 1916*. 1st ed. Carbondale, Ill.: Southern Illinois University Press, 1985. Print. <pdf available -<https://s3.amazonaws.com/arena-attachments/190319/2a5836b93124f200790476e08ecc4232.pdf>>

Dewey, John. My Pedagogic Creed. Dewey.pragmatism.org. N.p., 1897. Web. 6 Nov 2016 . <<http://dewey.pragmatism.org/creed.htm>>.

Dewey, John. On Education, Experience And Community. infed.org. N.p., 1997. Web. 12 Nov 2016 . <<http://infed.org/mobi/john-dewey-on-education-experience-and-community/>>.

Dictionary.com. *Definition of Scientific Method*. Dictionary.com. Web. 12 Nov 2016 . <<http://www.dictionary.com/browse/scientific-method>>.

Dictionary.com. *Definition Of Skill*. Dictionary.com. Web. 12 Nov 2016 . <<http://www.dictionary.com/browse/skill>>.

Dictionary.com. *Definition Of Wisdom*. Dictionary.com. Web. 12 Nov 2016 .
<<http://www.dictionary.com/browse/skill>>.

Dischler, Jerry. Building For The Next Moment. Inside AdWords. N.p., 2015. Web. 17 Nov 2016 .
<<https://adwords.googleblog.com/2015/05/building-for-next-moment.html>>.

Dragland, Åse. *Science News | Big Data, For Better Or Worse: 90% Of World's Data Generated Over Last Two Years*. Science Daily. N.p., 2012. Web. 12 Nov 2016 .
<<https://www.sciencedaily.com/releases/2013/05/130522085217.htm>>.

Duolingo: Learn Spanish, French And Other Languages For Free. Duolingo.com. N.p., 2016. Web. 12 Nov 2016 .
<<https://www.duolingo.com/info>>.

Eggers, William D. "Data As The New Currency". Deloitte University Press. N.p., 2013. Web. 12 Nov 2016 .
<<https://dupress.deloitte.com/dup-us-en/deloitte-review/issue-13/data-as-the-new-currency.html>>.

Ericsson.com. *Ericsson Mobility Report – On The Pulse of Networked Society*. Ericsson.com - Nov. 2015 Mobility Report. N.p., 2015. Web. 12 Nov 2016 . <<https://www.ericsson.com/mobility-report>>.

Ericsson.com. *Ericsson Traffic Exploration Tool*. Ericsson.com. N.p., 2016. Web. 3 Nov 2016 .
<<https://www.ericsson.com/TET/trafficView/loadBasicEditor.ericsson>>.

Facebook Inc. Facebook - Home. Investor.fb.com. N.p., 2016. Web. 15 Nov 2016 . <<http://investor.fb.com>>.

Faggella, Daniel. Venture Investments In Artificial Intelligence - Trends In 2016 And Beyond . TechEmergence.com. N.p., 2016. Web. 15 Nov 2016 . <<http://techemergence.com/investments-in-artificial-intelligence/>>.

Ford, Paul. "Are We Smart Enough To Control Artificial Intelligence?". MIT Technology Review. N.p., 2015. Web. 10 Oct. 2016 . <<https://www.technologyreview.com/s/534871/our-fear-of-artificial-intelligence/>>.

globaltopia. *TOP 10 [Global] CHALLENGES*. globaltopia.org N.p., 2016. Web. 20 Oct. 2016 .
<http://globaltopia.org/TOP_10_CHALLENGES.html>.

Google Cloud Platform. *Cloud Natural Language API*. Cloud.google.com. N.p., 2016. Web. 15 Nov 2016 .
<<https://cloud.google.com/natural-language/>>.

Harvard Medical School. *Life Expectancy Rises For The Educated; The Less-educated Reap No Benefit*. ScienceDaily. ScienceDaily, 12 March 2008. <www.sciencedaily.com/releases/2008/03/080311081149.htm>

Havel, John. *Recaptcha: The Genius Who's Tricking The World Into Doing His Work*. The Hustle. N.p., 2015. Web. 12 Nov 2016 . <<http://thehustle.co/the-genius-whos-tricking-the-world-into-doing-his-work-recaptcha>>.

Haverbeke, Marijn. *Eloquent Javascript. 1st ed*. San Francisco: No Starch Press, 2011. Print. <pdf
http://eloquentjavascript.net/Eloquent_JavaScript.pdf>

Hawking, Stephen. "Black Holes Ain't As Black As They Are Painted, Professor Stephen Hawking: Black Holes, The Reith Lectures - BBC Radio 4". *BBC*. N.p., 2016. Web. 9 Dec. 2016 .
<<http://www.bbc.co.uk/programmes/b06qjzv8>>.

Heimans, Jeremy and Henry Timms. *Understanding "New Power"*. Harvard Business Review. N.p., 2014. Web. 6 Nov 2016 . <<https://hbr.org/2014/12/understanding-new-power>>.

Howe, Neil. *Why Millennials Are Texting More And Talking Less*. Forbes.com. N.p., 2015. Web. 12 Nov 2016 .<<http://www.forbes.com/sites/neilhowe/2015/07/15/why-millennials-are-texting-more-and-talking-less/>>.

Inclusive Design Toolkit. *What Is Inclusive Design?*. InclusiveDesignToolkit.com. N.p., 2016. Web. 7 Nov 2016 .
<<http://www.inclusivedesigntoolkit.com/betterdesign2/whatis/whatis.html>>.

International Bureau of Education. *Jean Piaget*. Unesco.org. Web. 17 Oct. 2016 .
<<http://www.ibe.unesco.org/en/ibe-staff/jean-piaget>>.

International Business Machines - *What Is Big Data?*. ibm.com. N.p., 2016. Web. 12 Nov 2016 . <<https://www-01.ibm.com/software/data/bigdata/what-is-big-data.html>>.

International Data Corporation. *Developer Insights Report. 1st ed. Application Developer Alliance*, 2015. Web. 15 Nov 2016 .
<<https://static1.squarespace.com/static/53864718e4b07a1635424cdd/t/55de3d9ee4b07afa63281bf2/1440628126432/Apps+Alliance+Global+Developer+Insights+Report+2015.pdf>>.

International Montessori Index, *Montessori FAQ's for General, Teachers, Schools*. Montessori.edu. N.p., 2016. Web. 12 Nov 2016 . <<http://www.montessori.edu/FAQ.html>>.

Kaku, Michio. "Page 118". Future Physics. N.p., 2011 <<http://www.brautech.co.za/info/FuturePhysics.pdf>>.

Kerry, John. *Department of State & USAID Joint Strategy on Countering Violent Extremism*. N.p., May 2016. Web. 4 Nov. 2016 . <<http://www.state.gov/documents/organization/257913.pdf>>.

Khalaf, Simon. *Media, Productivity & Emojis Give Mobile Another Stunning Growth Year*. Flurry Blog. Web. 17 Nov 2016 . <<http://flurrymobile.tumblr.com/post/136677391508/stateofmobile2015>>.

Khan, Salman. *One World Schoolhouse, The: Education Reimagined*. 1st ed. New York: Hatchette Book Group, 2012. Print.

Ki-moon, Ban. *The United Nations Global Counter-Terrorism Strategy*. Un.org. Web. 24 Nov 2015.
<http://www.un.org/en/ga/search/view_doc.asp?symbol=A/70/674>

Kubrick, Stanley, Arthur C. Clarke, Keir Dullea, Gary Lockwood, William Sylvester, Dan Richter, Ray Lovejoy, Geoffrey Unsworth, Aram Khachaturi, György Ligeti, Johann Strauss, and Richard Strauss. 2001, a Space Odyssey. Burbank, CA: Warner Home Video, 2001.

Kurzweil, Ray. *The Law Of Accelerating Returns*, Kurzweilai.net. N.p., 2001. Web. 12 Nov 2016 .
<<http://www.kurzweilai.net/the-law-of-accelerating-returns>>.

Kurzweil, Ray. *The Singularity Is Near*. 1st ed. New York: Viking, 2005. Print.

LEGO. History - Mindstorms LEGO.Com. Lego.com. N.p., 2016. Web. 7 Nov. 2016 . <<https://www.lego.com/en-us/mindstorms/history>>.

Lella, Adam. Number Of Mobile-Only Internet *Users Now Exceeds Desktop-Only In The U.S.* ComScore, Inc. Web. 17 Nov 2016 . <<https://www.comscore.com/fre/Insights/Blog/Number-of-Mobile-Only-Internet-Users-Now-Exceeds-Desktop-Only-in-the-U.S>>.

Mandela, Nelson. Quoted by Un.org. N.p., 2014. Web. 12 Nov 2016. <<http://www.un.org/millenniumgoals/advocates/africarisingprogramme.shtml>>.

Manyika, James and Michael Chui. *Open Data: Unlocking Innovation And Performance With Liquid Information.* McKinsey & Company. N.p., 2013. Web. <<http://www.mckinsey.com/business-functions/digital-mckinsey/our-insights/open-data-unlocking-innovation-and-performance-with-liquid-information>>.

Mark, Joshua, Priscila Scoville, and Cristian Violatti. On Writing - Ancient History Encyclopedia. N.p., 2016. Web. 12 Nov 2016 . <<http://www.ancient.eu/writing/>>.

Marques, Juliana. *A Short History Of Moocs And Distance Learning*, MocoNewsAndReviews.com. N.p., 2013. Web. 12 Nov 2016 . <<http://moocnewsandreviews.com/a-short-history-of-moocs-and-distance-learning/>>.

McLeod, Saul. *Jean Piaget | Cognitive Theory*. SimplyPsychology.org. N.p., 2016. Web. 12 Nov 2016 . <<http://www.simplypsychology.org/piaget.html>>.

Merriam-webster. *Definition of Communicate*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/communicate>>.

Merriam-webster. *Definition Of Information*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/information>>.

Merriam-webster. *Definition Of Instruction*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/instruction>>.

Merriam-webster. *Definition Of Knowledge*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/knowledge>>.

Merriam-webster. *Definition Of Terrorism*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/terrorism>>.

Merriam-webster. *Definition Of Understanding*. Merriam-webster.com. Web. 12 Nov 2016 . <<https://www.merriam-webster.com/dictionary/understanding>>.

Merriam-Webster's Learner's Dictionary. *Definition of Learning for English-Language Learners From Merriam-Webster's Learner's Dictionary*. LearnersDictionary.com. N.p., 2014. Web. 12 Nov 2016 . <<http://www.learnersdictionary.com/definition/learning>>.

Metz, Cade. *Inside Openai, Elon Musk'S Wild Plan To Set Artificial Intelligence Free*. WIRED. N.p., 2016. Web. 5 Nov 2016 . <<https://www.wired.com/2016/04/openai-elon-musk-sam-altman-plan-to-set-artificial-intelligence-free/>>.

Michael Quoc, Zipfworks. *The State Of Bots: 11 Examples Of Conversational Commerce In 2016*. VentureBeat. N.p., 2016. Web. 15 Nov 2016 . <<http://venturebeat.com/2016/06/16/the-state-of-bots-11-examples-of-conversational-commerce-in-2016/>>.

Microsoft Corporation. *Design - Inclusive*. Microsoft.com. N.p., 2016. Web. 12 Nov 2016 . <<https://www.microsoft.com/en-us/design/practice/howwemake-section>>.

Microsoft Corporation. *The Market For Accessible Technology*. Microsoft.com. N.p., 2017. Web. 5 Nov. 2017 . <<https://www.microsoft.com/enable/research/phase1.aspx>>.

Millennium Alliance for Humanity and Biosphere. *About Page*. Stanford University. N.p., 2016. Web. 12 Nov 2016 . <[http:// mahb.stanford.edu/blog/the-challenge-for-humanity/](http://mahb.stanford.edu/blog/the-challenge-for-humanity/)>.

Montessori Education UK. *Montessori Media Center*. Montessori.org.uk. N.p., 2016. Web. 7 Nov 2016 . <http://www.montessori.org.uk/magazine-and-jobs/library_and_study_resources/teacher-training-study-resources/topics/peace>.

Montessori, Maria and Anne E. (translation by) George. *The Montessori Method - Translated*. Digital.library.upenn.edu. N.p., 1912. Web. 3 Nov. 2016 . <<http://digital.library.upenn.edu/women/montessori/method/method.html>>.

Montessori, Maria and Gerald Lee Gutek. *The Montessori Method. 1st ed*. Lanham, Md.: Rowman & Littlefield Publishers, 2004. Online PDF. <<http://www.arvindguptatoys.com/arvindgupta/mindstorms.pdf>>

Montessori, Maria. *Education and Peace*. 1943 Gerard Arthus. <https://archive.org/details/Peace_And_Education_>.

Mother Teresa Foundation. *About Page :: Mother Teresa Foundation*. Motherteresafoundation.org.in. N.p., 2010. Web. 12 Nov 2016 . <<http://www.motherteresafoundation.org.in/about>>.

Mother Teresa Foundation. *Welcome Page :: Mother Teresa Foundation*. Motherteresafoundation.org.in. N.p., 2010. Web. 12 Nov 2016 . <<http://www.motherteresafoundation.org.in/>>.

Mugridge, Rick. (Pages 1-3). *Test Driven Development and the Scientific Method*. Department of Computer Science, University of Auckland, NZ. N.p., 2003. Web. 8 Nov. 2016. <http://ftp://ftp.ufv.br/dpi/mestrado/TDD/Mugridge_TDD_Scientific_Method.pdf>.

Munro, A. Sandy. *History Of Lean Design® | Munro & Associates, Inc.*. Leandesign.com. Web. 17 Nov 2016 . <<http://leandesign.com/history-of-lean-design/>>.

Newlands, Murray and Read more. *The One Certainty About The Future Is The Pace Of Change Will Only Quicken*. Entrepreneur. N.p., 2015. Web. 5 Nov 2016 . <<https://www.entrepreneur.com/article/241255>>.

Newton, Richard. 2033: *Grave New World Or Brave New World? Two Visions Of Life And Society In 17 Years Time*. Richard-newton.com. N.p., 2017. Web. 6 Jan. 2017 . <<http://www.richard-newton.com/2033-grave-new-world-or-brave-new-world-two-visions-of-life-and-society-in-17-years-time/>>..

Noyes, Katherine. *10 Reasons Open Source Is Good For Business*. PCWorld. N.p., 2010. Web. 15 Nov 2016 . <http://www.pcworld.com/article/209891/10_reasons_open_source_is_good_for_business.html>

Obama, Barack. *Address To The United Nations General Assembly*. whitehouse.gov. N.p., 2014. Web. 8 Nov 2016 . <<http://www.whitehouse.gov/the-press-office/2014/09/24/remarks-president-obama-address-united-nations-general-assembly>>.

Olavsrud, Thor. *IT Wants (But Struggles) To Operationalize Big Data*. CIO. N.p., 2016. Web. 12 Nov 2016 . <<http://www.cio.com/article/3075423/it-strategy/it-wants-but-struggles-to-operationalize-big-data.html>>.

Open Data Institute. *What Is Open Data? | Open Data Institute*. N.p., 2016. Web. 15 Nov 2016 . <<http://theodi.org/what-is-open-data>>.

OpenAI. *Openai | About-Page*. OpenAI. N.p., 2016. Web. 15 Nov 2016 . <<https://openai.com/about/>>.

Ord, Ph.D. Toby. *Will We Cause Our Own Extinction? Natural Versus Anthropogenic Extinction Risks – CRASSH*. Crassh.cam.ac.uk. N.p., 2015. Web. 7 Nov 2016 . <<http://www.crassh.cam.ac.uk/gallery/audio/dr-toby-ord-will-we-cause-our-own-extinction-natural-versus-anthropogenic-e>>.

Oxford Dictionaries. *Definition Of Auto-Education In English*. Oxford Dictionaries | English. Web. 17 Nov 2016 . <<https://en.oxforddictionaries.com/definition/us/auto-education>>.

Oxford Dictionaries. *Definition Of Command In English* . Oxford Dictionaries Web. 12 Nov 2016 . <<https://en.oxforddictionaries.com/definition/command> >.

Oxford Dictionaries. *Definition Of Education In English | Oxford Dictionaries*. Web. 12 Nov 2016 . <<https://en.oxforddictionaries.com/definition/education>>.

Oxford Dictionaries. *Definition Of Epistemology In English | Oxford Dictionaries*. Oxford Dictionaries. Web. 12 Nov 2016 . <<https://en.oxforddictionaries.com/definition/education>>.

Oxford Dictionaries. *Definition Of Process In English | Oxford Dictionaries*. Web. 12 Nov 2016 . <<https://en.oxforddictionaries.com/definition/process>>.

Oxford Dictionaries. *Definition Of Singularity In English | Oxford Dictionaries*. oxforddictionaries.com. N.p., 2016. Web. 5 Nov 2016 . <<https://en.oxforddictionaries.com/definition/singularity>>.

Papert, Seymour. *Mindstorms: Children, Computers, and Powerful Ideas*. 1st ed. New York: Basic Books, 1980. Print. <<http://www.arvindguptatoys.com/arvindgupta/mindstorms.pdf>>

Pattison, Matthew and Alex Stedmon. *Inclusive Design And Human Factors: Designing Mobile Phones For Older Users*. N.p., 2006. Web. 12 Nov 2016 . <<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.446.5036&rep=rep1&type=pdf>>.

Pearcy, Mark. *The Allegory Of The Cave By Plato: Summary, Analysis & Explanation - Video & Lesson Transcript*. Study.com. Web. 12 Nov 2016 . <<http://study.com/academy/lesson/the-allegory-of-the-cave-by-plato-summary-analysis-explanation.html>>.

Phillips, D.C. and Harvey Siegel. *Philosophy Of Education*. Plato.stanford.edu. N.p., 2008. Web. 17 Nov 2016 . <<http://plato.stanford.edu/entries/education-philosophy/>>.

Plato. *Allegory of the cave*, Translated by Thomas Sheehan, Stanford professor of philosophy. Stanford.edu , 2000. Web. <<https://web.stanford.edu/class/ihum40/cave.pdf> >.

Plato.. Translated by Benjamin Jowett. *The Dialogues Of Plato. 1st ed*. New York: Random House, 1937. Print. <pdf <http://izt.ciens.ucv.ve/ecologia/Archivos/Filosofia-II/Plato--Laches.pdf>>.

Prashanth, Amitha. *Preschool With Montessori Education System, Prayag - Bangalore, India*. Prayag.in. Web. 6 Nov 2016 . <http://www.prayag.in/maria_montessori>.

Press, Gil. *A Very Short History of Big Data*, Forbes.com. N.p., 2013. Web. 12 Nov 2016 . <<http://www.forbes.com/sites/gilpress/2013/05/09/a-very-short-history-of-big-data/#4d6293b955da>>.

Prince, Michael. "Does Active Learning Work? A Review Of The Research". *Journal of Engineering Education* 93.3 (2004): 223-231. Web. <http://www4.ncsu.edu/unity/lockers/users/f/felder/public/Papers/Prince_AL.pdf>

Ries, Eric. *The Lean Startup | Methodology*. Theleanstartup.com. Web. 7 Nov 2016 . <<http://theleanstartup.com/principles>>.

Rosetta Stone. *About Page | A Language-Learning Company - Rosetta Stone®*. Rosettastone.eu. N.p., 2016. Web. 12 Nov 2016 . <<http://www.rosettastone.eu/about>>.

Ross, Alex (2016) *The Industries of the Future*. Available at: <http://www.audible.com/> (Downloaded: 19 March 2016).

Russell L. Ackoff. *Management Consultant & Systems Thinker*. Ackoffcenter.blogs.com. N.p., 2009. Web. 12 Nov 2016. <http://ackoffcenter.blogs.com/ackoff_center_weblog/2009/10/russell-l-ackoff-management-consultant-systems-thinker-90.html>.

Ryan, Ph.D. Donald. *History Of Writing*. *Historian.net*. N.p., 2016. Web. 8 Nov. 2016 . <<http://www.historian.net/hxwrite.htm>>.

Sample, Ian. *Most Threats To Humans Come From Science And Technology, Warns Hawking*. The Guardian. N.p., 2016. Web. 12 Nov 2016 . <<https://www.theguardian.com/science/2016/jan/19/stephen-hawking-warns-threats-to-humans-science-technology-bbc-reith-lecture>>.

Segaran, Toby. *Programming Collective Intelligence. 1st ed*. Sebastapol, Calif.: O'Reilly, 2007. Print.

Shuttleworth, Martyn. *What Is The Scientific Method?*. Explorable.com. N.p., 2016. Web. 5 Nov 2016 . <<https://explorable.com/what-is-the-scientific-method>>.

Simpson, Ronnie. *Mobile And Tablet Internet Usage Exceeds Desktop For First Time Worldwide* | Statcounter Global Stats. Statcounter.com. N.p., 2016. Web. 12 Nov 2016 . <<http://gs.statcounter.com/press/mobile-and-tablet-internet-usage-exceeds-desktop-for-first-time-worldwide>>.

Solomon, Jay and Carol E. Lee. *Obama Seeks To Fortify Iran Nuclear Deal*. WSJ. N.p., 2016. Web. 7 Nov 2016 . <<http://www.wsj.com/articles/obama-seeks-to-fortify-iran-nuclear-deal-1479692853>>.

Spector, Dina. *Here's How Many Days A Person Can Survive Without Water*. Business Insider. N.p., 2017. Web. 6 Oct. 2016 . <<http://www.businessinsider.com/how-many-days-can-you-survive-without-water-2014-5>>.

Stanford University. *About | The Stanford Natural Language Processing Group*. Nlp.stanford.edu. N.p., 2016. Web. 4 Nov 2016. <<http://nlp.stanford.edu/>>.

Stewart, James. *Facebook Has 50 Minutes Of Your Time Each Day. It Wants More*. Nytimes.com. N.p., 2016. Web. 17 Nov 2016 <<http://www.nytimes.com/2016/05/06/business/facebook-bends-the-rules-of-audience-engagement-to-its-advantage.html>>

Thayer-Bacon, Barbara. *In Knowledge There is Peace (Volume 5, number 3, 307-319)*. InfactisPax.org N.p., 2011. Web. 4 Nov 2016 . <<http://www.infactispax.org/volume5dot3/ThayerBacon.pdf>>.

The World Bank. Data.worldbank.org. N.p., 2016. Web. 12 Nov 2016 . <<http://data.worldbank.org/indicator/SP.DYN.LE00.IN>>.

Topol, Eric J. *The Patient Will See You Now: The Future of Medicine Is in Your Hands*. Basic Books, 2015. Print.

Tracxn Research - *Chatbots Startup Landscape*, June 2016. Tracxn Blog. N.p., 2016. Web. 12 Nov 2016 . <<https://blog.tracxn.com/2016/06/16/top-funded-startups-business-models-and-the-most-active-investors-in-chatbots/>>.

Trivedi, Deepika, Seema Bissa Harsh, and Rajesh Kumar. *A Brief Study On Machine Learning Techniques*. N.p., 2015. Web. 26 Nov. 2016. <http://ijiset.com/vol2/v2s1/IJISSET_V2_I1_05.pdf>.

Trivett, Vincent. *25 US Mega Corporations: Where They Rank If They Were Countries*. Business Insider. N.p., 2011. Web. 17 Nov 2016 . <<http://www.businessinsider.com/25-corporations-bigger-tan-countries-2011-6>>.

U.S Department of Health & Human Services. *User-Centered Design Basics*. Usability.gov. N.p., 2016. Web. Nov. 2016 . <<https://www.usability.gov/what-and-why/user-centered-design.html>>.

United Nations Educational, Scientific and Cultural Organization. UNESCO. N.p., 2016. Web. 12 Nov 2016 . <<http://en.unesco.org/>>.

United Nations. *UN News - General Assembly Declares Access To Clean Water And Sanitation Is A Human Right*. UN News Service Section. N.p., 2010. Web. 12 Nov 2016 . <<http://www.un.org/apps/news/story.asp?NewsID=35456#.WBa0l-ErLdQ>>.

United States Government. *User-Centered Design Basics*. Usability.gov. Web. 12 Nov 2016 . <<https://www.usability.gov/what-and-why/user-centered-design.html>>.

Usability First. *Introduction To User-Centered Design*. Usabilityfirst.com. N.p., 2016. Web. 12 Nov 2016 . <<http://www.usabilityfirst.com/about-usability/introduction-to-user-centered-design/>>.

Valentino-DeVries, Jennifer. "Steve Jobs'S Best Quotes". WSJ. N.p., 2011. Web. 10 Nov. 2016 . <<http://blogs.wsj.com/digits/2011/08/24/steve-jobss-best-quotes/>>.

Vesselinov, Roumen and John, Grego. *Duolingo 'Effectiveness' Study* N.p., 2012. Web. 12 Nov 2016 . <<https://s3.amazonaws.com/duolingo-papers/other/vesselinov-grego.duolingo12.pdf>>.

Vitsoe. *Good Design | About* . Vitsoe.com. N.p., 2016. Web. 12 Nov 2016 . <<https://www.vitsoe.com/gb/about/good-design>>.

Walker, Patrick Gordon. *The Origins Of The Machine Age | History Today*. Historytoday.com. N.p., 1966. Web. 15 Nov 2016 . <<http://www.historytoday.com/patrick-gordon-walker/origins-machine-age>>.

Wasik, Bill. *#Riot: Self-Organized, Hyper-Networked Revolts—Coming To A City Near You*. WIRED. N.p., 2011. Web. 11 Oct. 2016 . <https://www.wired.com/2011/12/ff_riots/>.

Waters, Kelly. *What Is Agile? (10 Key Principles Of Agile) | All About Agile*. Allaboutagile.com. N.p., 2016. Web. 12 Nov 2016 . <<http://www.allaboutagile.com/what-is-agile-10-key-principles/>>.

Wells, H. G. *Adapt Or Perish, Now As Ever, Is Nature's Inexorable Imperative*. - *Quote Of The Day - English - The Free Dictionary Language Forums*. Forum.thefreedictionary.com. Web. 17 Nov. 2016 . <http://forum.thefreedictionary.com/postst56991_Adapt-or-perish--now-as-ever--is-nature-s-inexorable-imperative.aspx>.

Woods, Viveca. Gartner Inc. Newsroom, gartner.com Egham, UK. Feb 2016. Web. 12 Nov 2016 . <<http://www.gartner.com/newsroom/id/3215217>>. "Gartner Says Worldwide Smartphone Sales Grew 9.7 Percent in Fourth Quarter of 2015"

World Economic Forum. *The Transformative Power Of Technology - Digital Transformation of Industries*. World Economic Forum. Web. 7 Nov 2016 . <<http://reports.weforum.org/digital-transformation-of-industries/onward-and-upward-the-transformative-power-of-technology/>>.

Your Dictionary Online. Definition of Technology From Your Dictionary Online. Yourdictionary.com. Web. 12 Nov 2016 . <<http://www.yourdictionary.com/technology>>.